AF-400 INVERTER DRIVES

Standard Driver - 331X440AAG01

OPERATION – TROUBLE SHOOTING – REPAIR

GEK-62515

DRIVER ASSEMBLY	ELEM	36D868888AA
	CONN	36B590234AA

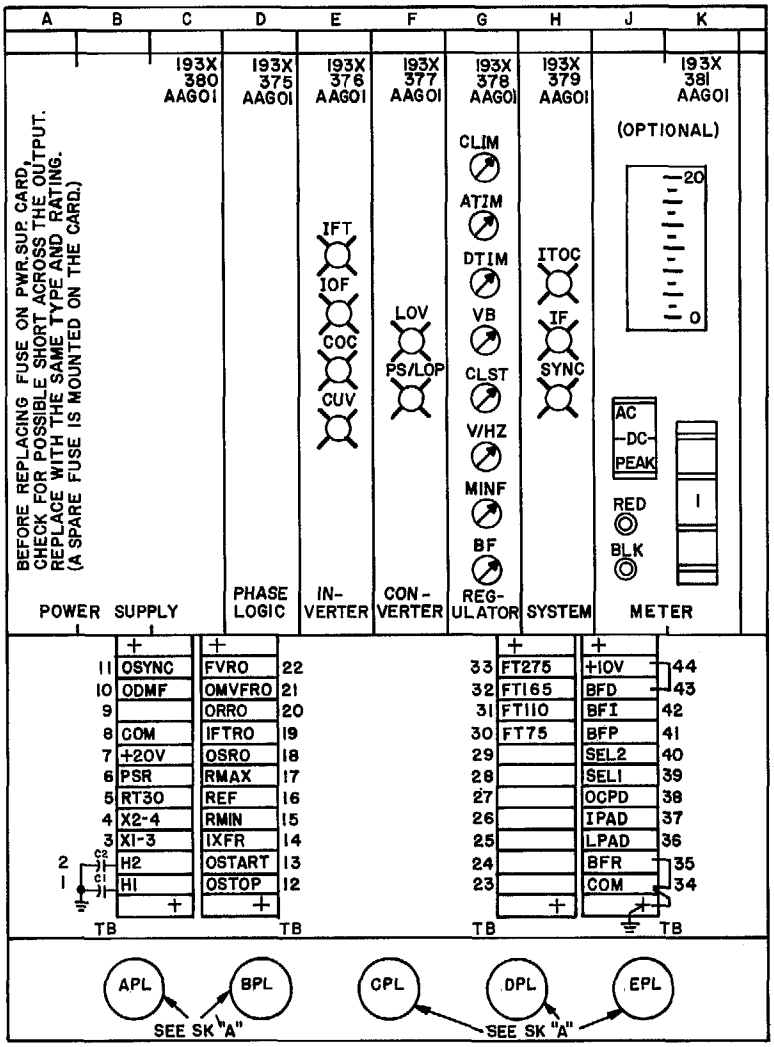
PRINTED CIRCUIT CARD DIAGRAMS

PHASE LOGIC	36D868807AA
INVERTER	36D868810AA
CONVERTER	36D868812AA
REGULATOR	36D868816AA
SYSTEM	36D868818AA
POWER SUPPLY	36C764164AA
	36D868820AA
METER	36D868822AA
CURRENT FEEDBACK	36C764166AA
PULSE TRANSFORMER	36C764172AA

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

demark of the General Electric Co.

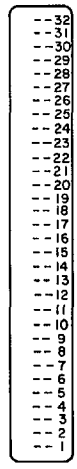


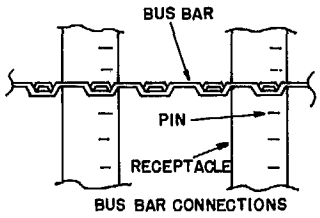
SH. NO. 1
36B590234AA
CONT. ON SH. 2
REV. NO. 0

ON PRINTED CIRCUIT CARDS USED IN THIS RACK THE LETTERS "AA" AFTER BASIC CATALOG NUMBER INDICATES ORIGINAL DESIGN. SUBSEQUENT DESIGNS WITH THE SAME BASIC NUMBERS AND GROUP NUMBER WITH THE SECOND LETTER CHANGED, SUCH AS AB, AC, AD, ETC., ARE DIRECTLY INTERCHANGEABLE AND MAY BE SUPPLIED IN PLACE OF THE "AA" CARDS.

THE PRINTED CIRCUIT CARD SHOULD ALWAYS BE REMOVED WITH THE CARD EXTRACTOR WHICH IS ATTACHED ON TOP OF THE CARD RACK. SOME CARDS CONTAIN PARTS WHICH WILL BE THERMALLY HOT AFTER BEING IN OPERATION. CARE SHOULD BE EXERCISED IN HANDLING ALL CARDS AFTER REMOVAL UNTIL THESE PARTS HAVE COOLED.

FRONT VIEW OF 32 PIN RECEPTACLE AS SEEN IN RACK CLOSED POSITION.

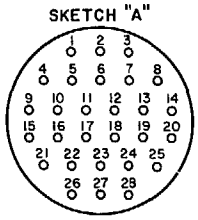




BUS BAR CONNECTIONS

FACTORY NOTES:
1. ASSEMBLE BUS BAR BEFORE WIRING. ALWAYS ASSEMBLE WITH THE BODY OF THE BUS TOWARD THE LOWER NUMBERED SIDE OF THE RECEPTACLE

SKETCH "A"



FRONT VIEW FROM PLUG SIDE

SYMBOLS

- POT ADJUSTMENT
- INDICATING LIGHT

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV. 3	REV. 2 m.a.th SV872 #28	REV. 1 MARCH 23, 1977 ETI\Wjw SV872 #IC	APPROVALS	GENERAL ELECTRIC	CONNECTION DIAGRAM AF-400 DRIVER	
REV. 4	ISSUED J.O'Hlayak MAR 3, 1977	PRINTS TO JA(GD) 5(S)	FIRST MADE FOR REQ. AF-400	SPEED VARIATOR PROD DEPT ERIE, PA., U.S.A.	36B590234AA CONT. ON SH. 2	SH. NO. 1
REV. 5	MADE BY B.Wisniewski Aug 27, 1976 50(BK) 575(50) 59C(B)	58(3) 23 59C(BK) 5	331X440AAG01			

BUS BAR TABLE		
TABS	LOCATION	CIRCUIT
01	C D E F G H K	+20V
02	C D E F G H K	COM
03	F G H K	+10V

METAL JUMPER TABLE
TB34 TO TB MTG. SCREW
TB34 TO TB 35
TB43 TO TB44

BACK-PLANE WIRING
#24 WHITE
C01-C31
C32-E05
D20-E31
D21-E16
D22-E32
D23-E19
D24-E26
D25-E27-F11-K06
D27-H24
D28-E10-F32-K07
D29-E17
D30-E18-K05
D31-H30-K09
D32-E28-H08-H09
E11-K32
E12-K31
E13-K30
E14-F04-H23
E15-F07
E20-G18-K19
E21-F15-G22-H26-K16
E29-F09
E30-F06
F08-G20-H27
F12-H10
F18-G24
F21-G19-K15
F25-G21-K14
F13-H13
G05-K17
G06-H31
G07-G23
G10-H04
G14-H05
G15-H07-K10
G16-H11-K08
G17-H12
G26-K11
G27-K12
G28-K1[illegible]
G29-H1[illegible]
G30-K18
G32-H28
G04-H16
G25-H14

PLUG WIRING
#24 WHITE
APL01-C10
APL02-D09
APL03-C03
APL05-C10
APL06-D08
APL07-C03
APL13-G28
APL14-H02
APL19-E11
APL20-E02
APL22-C10
APL23-D06
APL24-C03
APL26-C11
APL27-D05
APL28-C04
BPL01-C11
BPL02-D17
BPL03-C04
BPL05-C11
BPL06-D18
BPL07-C04
BPL13-G27
BPL14-G02
BPL19-E12
BPL20-D02
BPL22-C12
BPL23-D11
BPL24-C05
BPL26-C12
BPL27-D10
BPL28-C05
CPL01-C12
CPL02-D15
CPL03-C05
CPL05-C13
CPL06-D16
CPL07-C06
CPL13-G26
CPL14-F02
CPL19-E13
CPL20-D02
CPL22-C13
CPL23-D13
CPL24-C06
CPL26-C13
CPL27-D12
CPL28-C06
DPL01-C14
DPL02-F27
DPL03-C07
DPL04-C14
DPL05-F26
DPL06-C07
DPL09-C14
DPL10-F29

PLUG WIRING
#24 WHITE
DPL11-C07
DPL15-C15
DPL16-F28
DPL17-C08
DPL21-C15
DPL22-F31
DPL23-C08
DPL26-C15
DPL27-F30
DPL28-C08
EPL01-C16
EPL02-D07
EPL03-C09
EPL04-C16
EPL05-D19
EPL06-C09
EPL09-C16
EPL10-D14
EPL11-C09
EPL22-E04
EPL23-E02
EPL19-F20
EPL20-F19
EPL24-F16
EPL25-F17
EPL26-F24
EPL27-F22
EPL28-F23

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV. 3	REV. 2 [illegible] 8/1/72 M.C. Getty SV 872 # 11F	REV. 1 MARCH 23, 1977 W. Wisniewski SV 872 #11C	APPROVALS [illegible]	GENERAL ELECTRIC	AF-400 DRIVER WIRE TABLE
REV. 4	ISSUED J. P. Holzapfel MAR. 8, 1977	PRINTS TO 5(S) 3A(CD)	FIRST MADE FOR REQ. AF-400	SPEED VARIATOR DEPT. ERIE, PA. U.S.A.	36B590234AA SH. NO. 2
REV. 5	MADE BY W. Wisniewski Aug 30, 1977 5D(BK) 5T5(SO)	5B(3)23 5QC(BK)5 5QC(B)	331X440AAG01		CONT ON SH. 3

SV-167(5-69)

TERM. BD. WIRING
#24 WHITE
TB03-C27
TB04-C29
TB05-C21
TB06-C25
TB07-C01
TB08-C02
TB10-H32
TB11-H25
TB12-H21
TB13-H20
TB14-H22
TB15-H02
TB16-G30
TB17-*
TB18-H06
TB19-H29
TB20-H18
TB21-H19
TB22-G05

TERM. BD. WIRING
#24 WHITE
TB30-E22
TB31-E25
TB32-E24
TB33-E23
TB34-G02
TB35-G13
TB36-*
TB37-*
TB38-*
TB39-*
TB40-*
TB41-G09
TB42-G11
TB43-G12
TB44-G03

WIRE TABLE

1TX LEADS

LEAD	CONNECTION
(BLK)H1	TB01
(BLK)H2	TB02
(BRN)X1	TB03
(BRN)X2	TB04
(GRN)X3	TB03
(GRN)X4	TB04
(YEL)X5	CUT & INSULATE
(YEL)X6	CUT & INSULATE

C1 & C2 CONNECTIONS

LEAD	CONNECTION
C1-H1	TB01
C2-H2	TB02
C1,C2-GRD	TB MTG. SCREW

*CONNECT RESISTOR WIRES AS INDICATED BELOW:

RESISTOR NOMEN.	1ST END CONN.	2ND END CONN.
RMAX	TB17	H01
LPAD	TB36	F12
IPAD	TB37	H09
OCPD	TB38	K19
SEL1	TB39	K13(MOVEABLE END)
SEL2	TB40	K15(MOVEABLE END)

MOVEABLE JUMPER CONNECTIONS

THE FIRST WIRE CONNECTION IS THE PERMANENT ONE.
THE SECOND WIRE CONNECTION IS THE MOVEABLE ONE.

#24 RED

K28- K20

RESISTOR WIRES

SEL1 - } SEE RESISTOR WIRE TABLE
SEL2 - }

REV. 3	REV. 2	REV. 1	APPROVALS	GENERAL ELECTRIC	AF-400 DRIVER WIRE TABLE
REV. 4	ISSUED	PRINTS TO	FIRST MADE FOR REQ. AF-400		368590234AA SH. NO. 3
REV. 5	MADE BY	5B(3)23 5(QC)BK(5) 5TS(50) 5QC(D)	331X440AAG01		CONT. ON SH. 4

368590234AA SH. NO. 3 CONT. ON SH. 4

IF ONLY THE SECOND LETTER HAS CHANGED IN THE TWO LETTER GROUP OF THE CARD 193X NO., THE CARDS ARE INTERCHANGEABLE (AB REPLACES AA).

DRIVER

C	D	E	F	G	H	K
193X380AAG01	193X375AAG01	193X376AAG01	193X377AAG01	193X378AAG01	193X379AAG01	193X381AAG01
BEFORE REPLACING FUSE ON POWER SUPPLY CARD, CHECK FOR POSSIBLE SHORT ACROSS THE OUTPUT. REPLACE WITH THE SAME TYPE AND RATING FUSE. (A SPARE FUSE IS MOUNTED ON THE CARD.)		INVERTER FAULT TRIP INVERTER OVERFREQUENCY COMMUTATION OVERCURRENT CONTROL UNDERVOLTAGE	D.C. LINK OVERVOLTAGE WRONG PH. SEQ. LOSS OF PHASE	CURRENT LIMIT ACCEL. TIME DECEL. TIME VOLTAGE BOOST CURR LIM. STAB. VOLTS/HERTZ MINIMUM FREQ. BASE FREQ.	INVERSE TIME OVERCURRENT INVERTER FREQUENCY INVERTER SYNCHRONIZED	20 … 0 AC DC PEAK RED BLK I (OPTIONAL) METER
POWER SUPPLY	PHASE LOGIC	INVERTER	CONVERTER	REGULATOR	SYSTEM	

INPUTS	11	OSYNC	FVRO	22	READOUTS	OVER-FREQUENCY SELECTION	33	FT275	+10V 44
	10	ODMF	OMVFRO	21			32	FT165	BFD 43
	9		ORRO	20			31	FT110	BFI 42
	8	COM	IFTRO	19			30	FT75	BFP 41
	7	+20V	OSRO	18			29		SEL2 40
POWER SUPPLY	6	PSR	RMAX	17			28		SEL1 39
	5	RT30	REF	16			27		OCPD 38
	4	X2-4	RMIN	15	INPUTS		26		IPAD 37
	3	X1-3	IXFR	14			25		LPAD 36
115VAC INPUT	2	H2	OSTART	13			24		BFR 35
	1	H1	OSTOP	12			23		COM 34

EXT. BASE FREQUENCY POT. CONN.
DIAGNOSTIC
BASE FREQ. RANGE SEL.

15 1/2"

CARD REPLACEMENT INSTR.

MAKE POT SETTINGS (INCL. VLIM POT ON REGUL CARD) AND CARD JUMPERS SAME AS ON ORIGINAL CARD.

SPACE FOR CAUTION LABEL NP104X905BA148

460V JUMPER ON INVERTER CARD

460 VOLT INVERTER - JUMPER NEEDED
230 VOLT INVERTER - REMOVE JUMPER

60HZ JUMPER ON CONVERTER CARD

60 HZ AC SUPPLY - JUMPER NEEDED
50 HZ AC SUPPLY - REMOVE JUMPER

SPACE FOR CAUTION LABEL NP104X905BA067

BASE FREQUENCY RANGE SELECTION

BASE FREQ RANGE	37.5 TO 75HZ	75 TO 150HZ
JUMPER TB35 TO	TB34	TB35

FREQUENCY TRIP SELECTION

FREQUENCY TRIP	75HZ	110HZ	165HZ	275HZ
JUMPER TB30 TO	TB30	TB31	TB32	TB33

SPACE FOR WARNING LABEL NP104X905BA068

SELECTOR SWITCH POS.	OPTIONAL METER CARD SELECTED SIGNAL	SCALE SELECT.	METER READING (± 1.0 MAX ERROR) OFF CONDITION	0 REF. 0 LOAD	1/2 REF. 1/2 LOAD	1 REF. 1 LOAD	SPECIAL CONDITION
1	+10V - POWER SUPPLY MIDPOINT	DC	10	10	10	10	
2	REF - REFERENCE VOLTAGE	DC	①	0	7.5	15	
3	FVRO - INVERTER FREQUENCY	DC	0.3 TO 1.5	0.3 TO 1.5	6 ②	12 ②	
4	LVF - DC LINK VOLTAGE	DC	0 TO 1 5	0 TO 0.5	7.5 ③	15 ③	
5	CVF - CONVERTER VOLTAGE FEEDBACK	DC	9.7	10	7.7	5.5	
6	CVR - CONVERTER VOLTAGE REFERENCE	DC	10.5	8 TO 10 5	6	3.5	
7	CFA - AC MOTOR CURRENT - PHASE A	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
8	CFB - AC MOTOR CURRENT - PHASE B	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
9	CFC - AC MOTOR CURRENT - PHASE C	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
10	OCL - CURRENT LIMIT SIGNAL	DC	15 TO 17	15 TO 17	15 TO 17	15 TO 17	4 TO 8 AT CURR LIMIT
11	IRI - INVERTER RUN SIGNAL	DC	0	17 TO 19	17 TO 19	17 TO 19	
12	OIS - INVERTER SYNCHRONIZE SIGNAL	DC	17 TO 19	17 TO 19	17 TO 19	17 TO 19	0 AT SYNCH.
13	DFS - DELAYED FIRING SUPPLY	DC	17 TO 19	17 TO 19	17 TO 19	17 TO 19	
14	IIPT - INVERTER FIRING PULSE TRAIN	PEAK/DC	17 TO 19/1	17 TO 19/1	17 TO 19/1	17 TO 19/1	
15	ICFF - INVERTER 3 PHASE LOGIC SIGNAL	PEAK/DC	17 TO 19/0	17 TO 19/0	17 TO 19/0	17 TO 19/0	
16	CCA - INV. PEAK COMM. CURR. - PH. A	PEAK	0	9 TO 11	11 TO 13	12 TO 14	
17	CCB - INV. PEAK COMM. CURR. - PH. B	PEAK	0	9 TO 11	11 TO 13	12 TO 14	
18	CCC - INV. PEAK COMM. CURR. - PH. C	PEAK	0	9 TO 11	11 TO 13	12 TO 14	
19 ④	SEL - D TAB 5, 6, 7, 10, 11, 12, 13, 14 & 19	PEAK/DC	0	15 TO 18/0	15 TO 18/0	15 TO 18/0	0 AFTER FAULT
19 ④	SEL - D TAB 8, 9, 15, 16, 17 & 18	PEAK/DC	0	16 TO 19/2	16 TO 19/2	16 TO 19/2	0 AFTER FAULT
19 ④	SEL - F TAB 26, 27, 28, 29, 30 & 31	PEAK/DC	16 TO 19/1	16 TO 19/1	16 TO 19/1	16 TO 19/1	0 AFTER FAULT

① DEPENDENT ON REFERENCE SETTING.

② READINGS SHOWN ARE FOR 60HZ BASE FREQUENCY. ACTUAL FREQUENCIES ARE 5 TIMES METER READING FOR TAB35 CONN. TO TAB34 OR 10 TIMES METER READING FOR TAB35 OPEN.

③ ACTUAL DC LINK VOLTAGES ARE 20 TIMES METER READING FOR 230 VOLT DRIVES AND 40 TIMES METER READING FOR 460 VOLT DRIVES.

④ POS. 19 IS FOR DRIVER BACKPLANE SELECTOR PROBE. READINGS SHOWN ARE SCR FIRING SIGNALS.

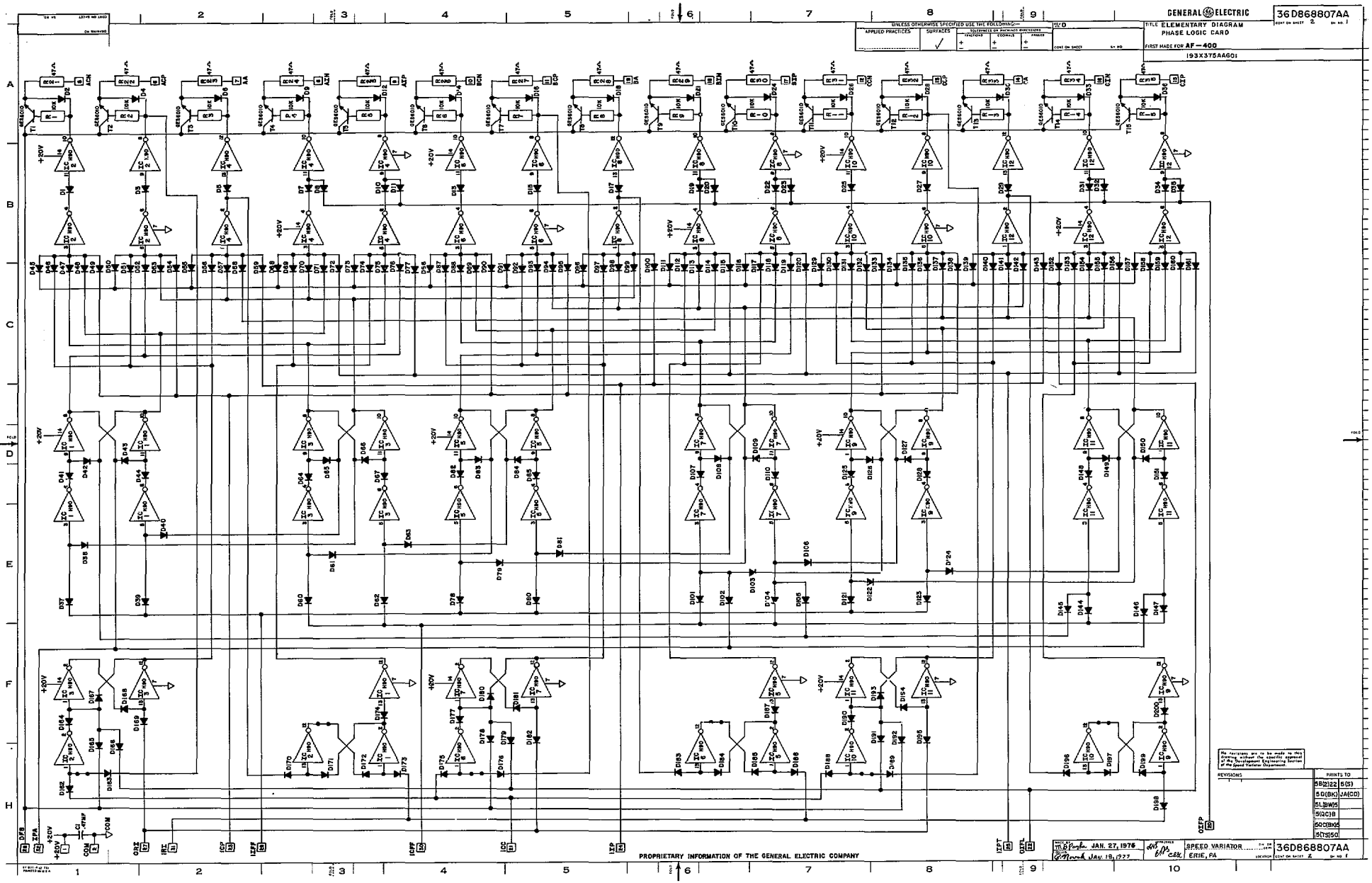
36B590 234AA SH.4

10"

1. SIZE AS INDICATED.
2 CUT TO SIZE

3. PRINT ON FASSON WHITE CONVENTION BADGE (LAPEL) STOCK NO 65P.

REV.		ISSUED		PRINTS TO BE MADE FROM	FIRST MADE FOR REQ	GENERAL ELECTRIC	LABEL ARTWORK AF-400 DRIVER	
REV. 1	MAR 3, 1977 SV872#11C	J.J. Volpp	MAR 3, 1977	5161 (BK) 5161(BK) 5161(BK) 5161(BK)	AF-400	SPEED VARIATOR PROD DEPT ERIE, PA., U.S.A.	36B590 234AA CONT ON SH. FL	SH. NO 4
REV. 2	SV872#28	MADE BY J.J. Volpp	FEB 7, 1977		331X440AAG01			



GENERAL ELECTRIC

36D868807AA

TITLE ELEMENTARY DIAGRAM
PHASE LOGIC CARD

FIRST MADE FOR AF-400

193X375AAG01

SPEED VARIATOR — ERIE, PA

36D868807AA

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS	

PRINTS TO	
5B(2)22	5(S)
5D(BK)	JA(CD)
5L(BW5	
5(QC)8	
5QC(BK5	
5(T)150	

GENERAL ELECTRIC 36D868807AA

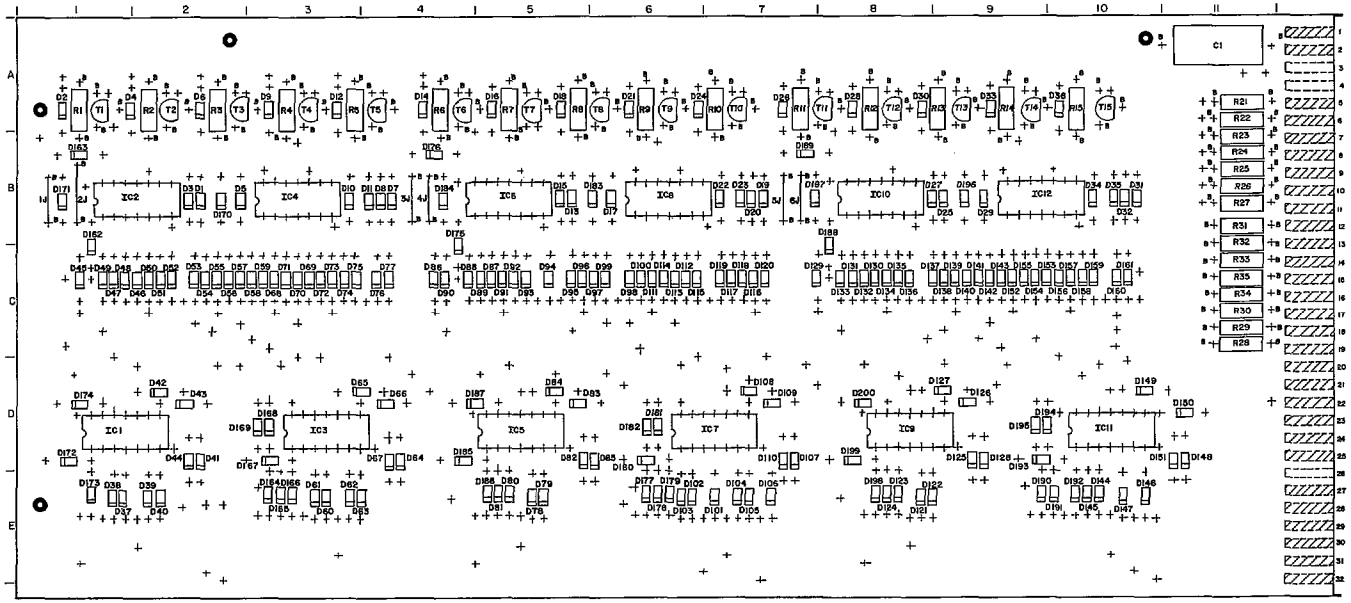
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES: SURFACES: √ TOLERANCES OF MACHINED DIMENSIONS: FRACTIONS ± DECIMALS ± ANGLES ±

REV 0

TITLE PRINTED CIRCUIT DIAGRAM PHASE LOGIC GARD

FIRST MADE FOR AF-400

193X375AA001



HOLE TABULATION

ALL HOLES ARE .032 EXCEPT THE HOLES TABULATED BELOW.

LOC	DIA.	QTY
A	.157	4
B	.040	113

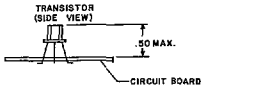
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $11.50^{+.000}_{-.020} \times 5.130^{+.000}_{-.020}$

TRANSISTOR LEAD SKETCH



COMPONENT HEIGHT SKETCH



2X	368605228AA005	368605229AA005
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.

REVISIONS

We certify we are so ready to this drawing without the specific approval of the Development Engineering Section of the Servo Variator Department.

PRINTS TO

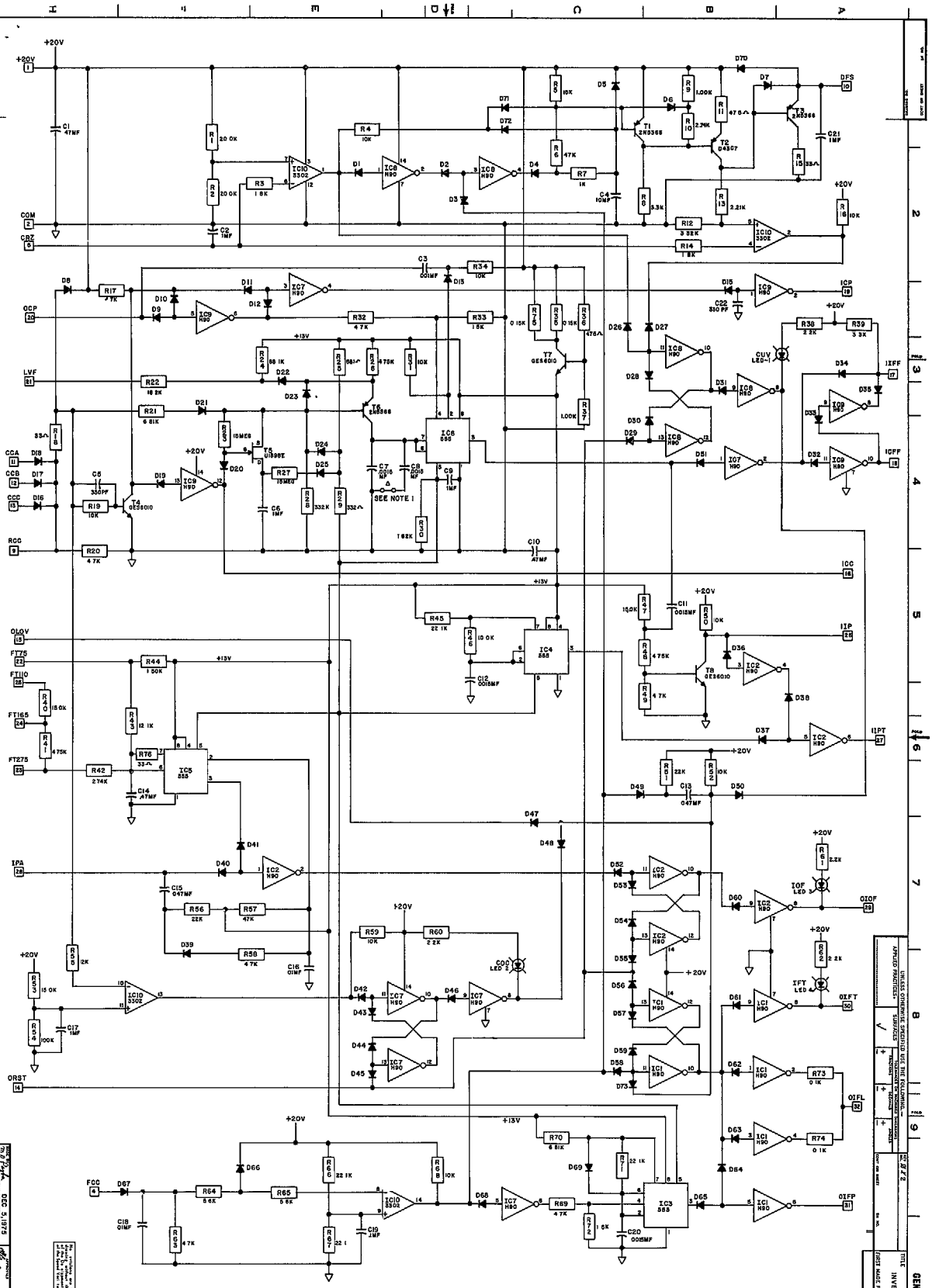
592(22	SS
5D(BK)	AW(BW)
5L(BW)5	JA(CD)
5(QC)B	DS(REP)
50C(2)K5	
5(TS)50	

DRAWN E. Marsh JAN. 23,1976 CHECKED [illegible] ENGINEER [illegible] APPROVED [illegible]

SPEED VARIATOR ERIE, PA.

36D868807AA

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY



NOTES
1 THIS JUMPER SHOULD BE PRESENT ON 460 VOLT DRIVES FOR PROPER OPERATION.
REMOVE THIS JUMPER FOR 230 VOLT DRIVES

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC
TITLE
INVERTER CARD
FIRST MADE FOR AF-400
193X376AAG01
36D86898 10AA 1

36D86881 10AA 1
SPEED VARIATOR
ERIE, PA
DRAWN
DEC 5 1975
APPROVED
APPROVED PRACTICES
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING
SURFACES
TREATMENT & FINISH
MATERIAL
TOLERANCES
DECIMALS
FRACTIONS
ANGLES
NEXT ASSY
USED ON
APPLICATION
DO NOT SCALE
SHEET 2 OF 2
REVISIONS
PRINTED TO

GENERAL ELECTRIC

36D868810AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES— | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
INVERTER CARD

FIRST MADE FOR AF-400

193X376AA001

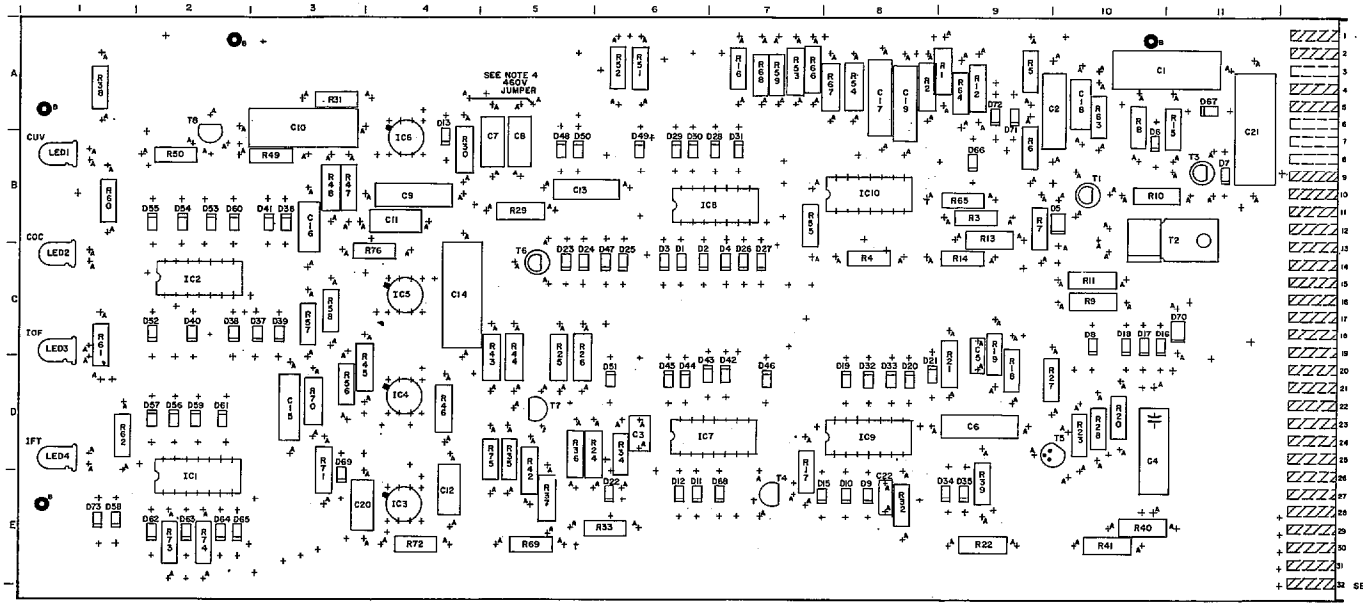
HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW

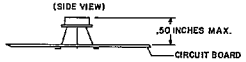
A — .040 — 234
B — .157 — 4

NOTES

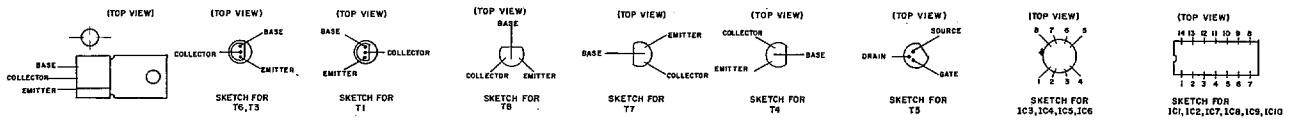
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED
3. CARD SIZE, 11.50 +.000 −.010 X 5.130 +.000 −.005
4. 460V JUMPER IS TO BE PUT ON ALL CARDS. THIS JUMPER WILL BE REMOVED FOR 230V APPLICATIONS.
5. ALL COMPONENTS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.



COMPONENT HEIGHT SKETCH
TRANSISTOR
(SIDE VIEW)



TRANSISTOR LEAD SKETCHES



1	0	2X	36B860523AA005	36B860523AA005
FRONT	BACK	SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.
SPACE UNIT REQUIREMENT				

REVISIONS

No revisions are to be made to this drawing without the written approval of the Development Engineering Section of the Speed Variator Department.

1	Engrg/RTP 12-6-76	CHKD. PER. 81-975#2&3	32
2	MAR. 2, 1978 Engrg/RTP	CHK. PER. 81-975#2&3	32

PRINTS TO

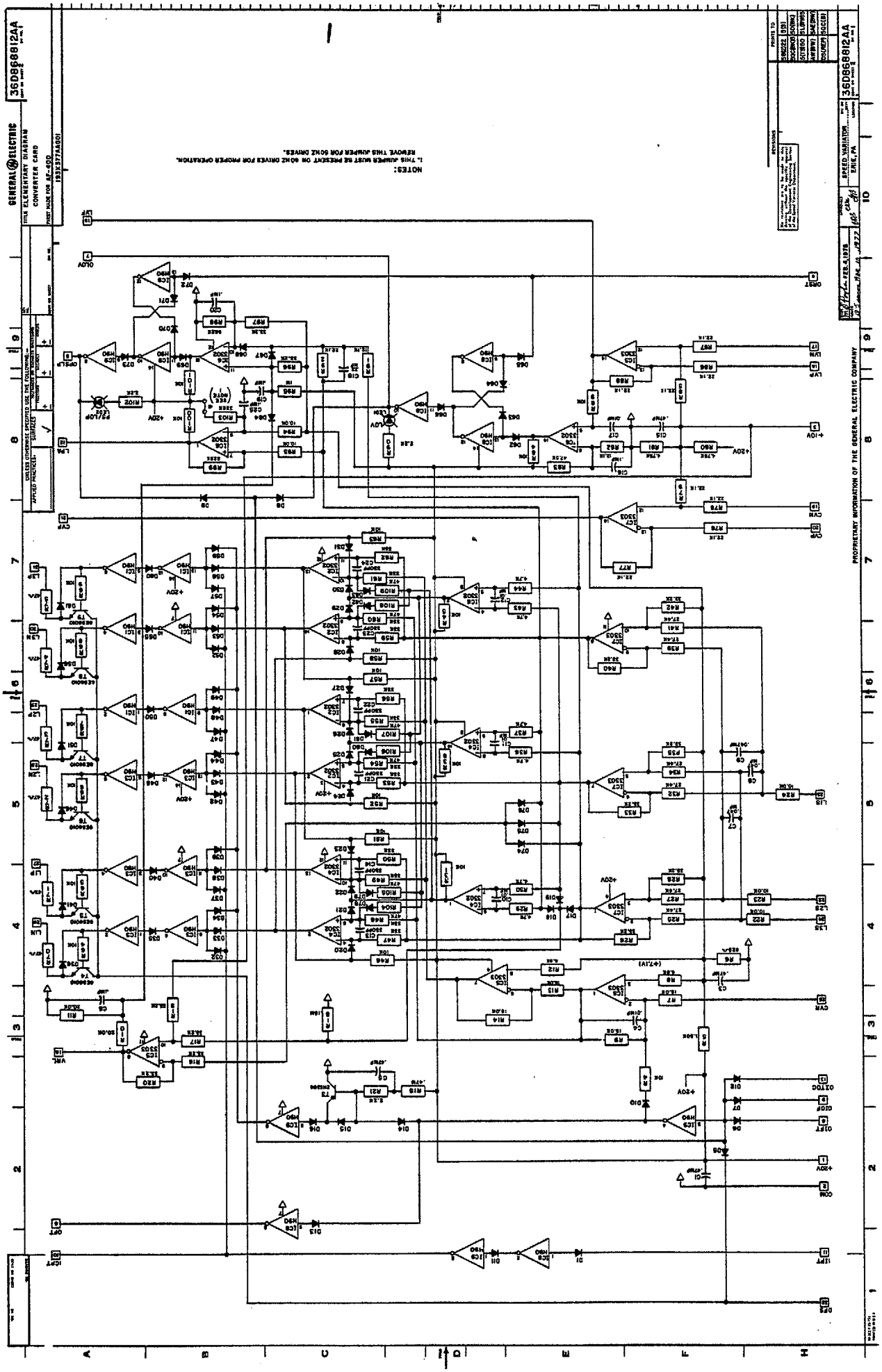
SB(B)M	3J(BW)
SR(2)BW	TS(50)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN M.E. Park FEB. 3, 1976
CHECKED EBS eek
APPROVED [illegible] APR 10, 1977

SPEED VARIATOR
ERIE, PA

36D868810AA


GENERAL ELECTRIC
36D86881 2AA
TITLE ELEMENTARY DIAGRAM
CONVERTER CARD
FIRST MADE FOR AC-600
193X377AA001
NOTES:
1. THIS JUMPER MUST BE PRESENT ON 60 HZ DRIVES FOR PROPER OPERATION.
REMOVE THIS JUMPER FOR 50 HZ DRIVES.
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATION
ERIE, PA
36D86881 2AA

GENERAL ELECTRIC

36D868812AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES: SURFACES ✓

TITLE PRINTED CIRCUIT DIAGRAM
CONVERTER CARD

FIRST MADE FOR AF-400

193X577AA001

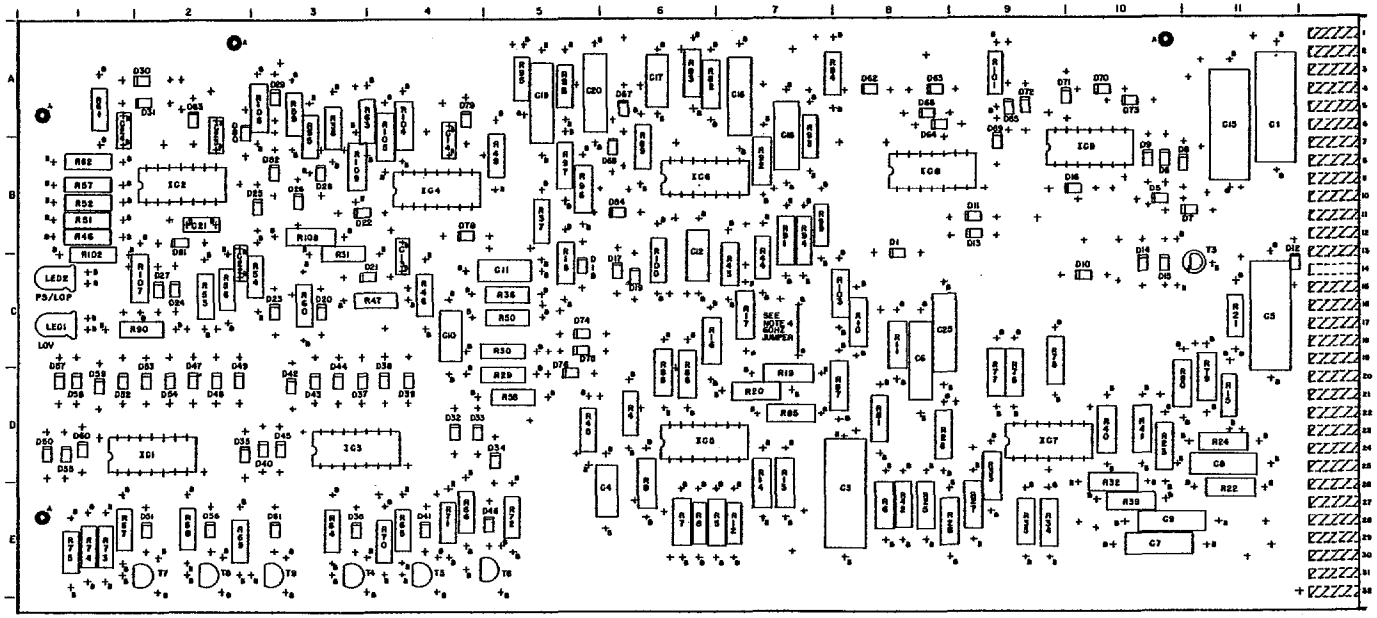
HOLE TABULATION

ALL HOLES ARE .032 DIA. EXCEPT THE HOLES TABULATED BELOW.

LOC.	DIA.	QTY.
A	.157	4
B	.040	287

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, 11.50 +.000 −.015 × 5.130 +.005 −.006
4. 60HZ JUMPER IS TO BE PUT ON ALL CARDS. THIS JUMPER WILL BE REMOVED FOR 50HZ APPLICATIONS.



TRANSISTOR LEAD SKETCHES



2X	368605237AA005	368605238AA005
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.

REVISIONS

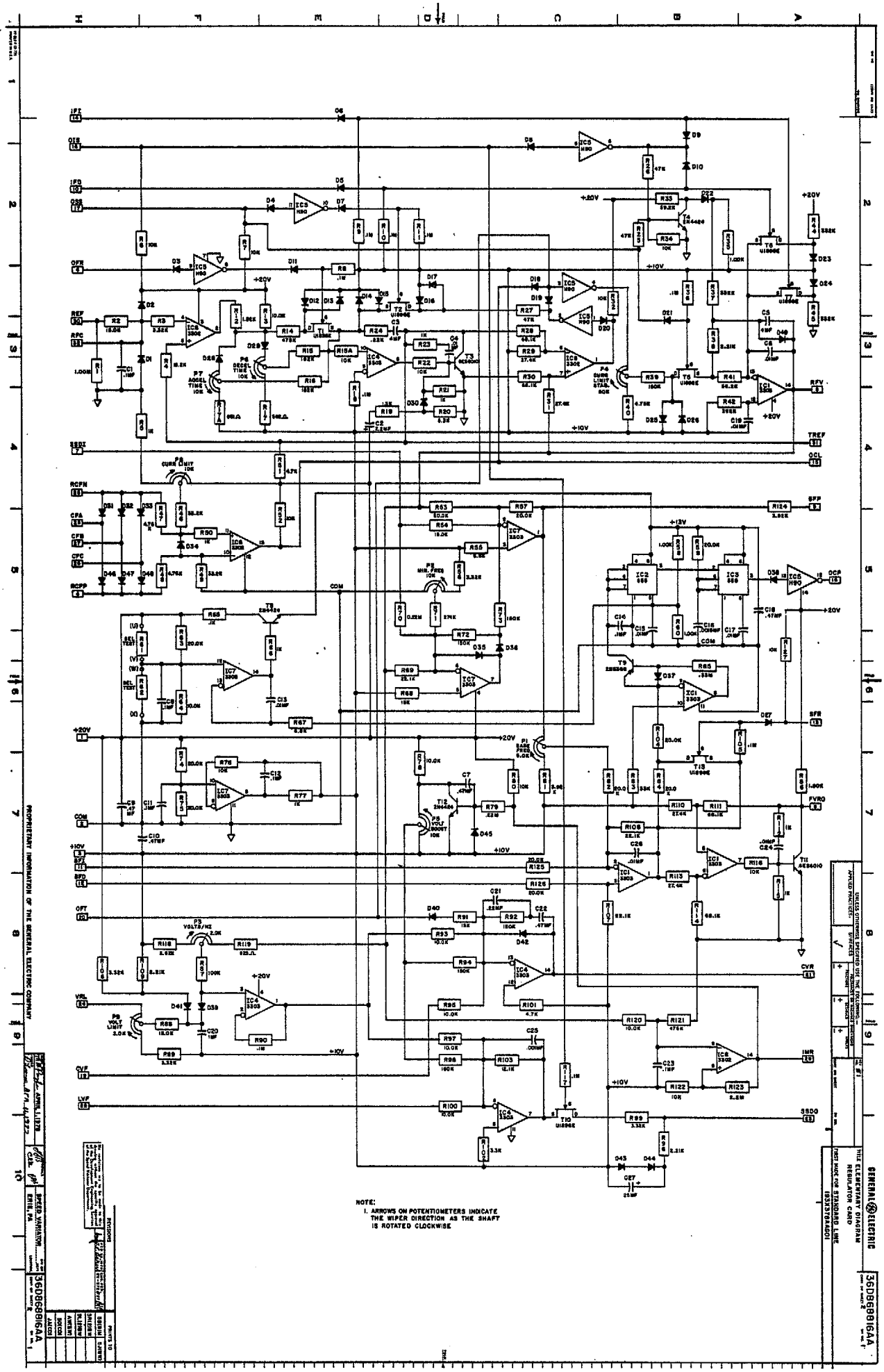
1	ECO J. HITE	DEC. 23, 1976		
	ECO C. KAISER	DEC. 27, 1976		

PRINTS TO

SB(BW)	SJ(BW)
SR(2BW)	TR(BO)
SL(2BW)	
AW(BW)	
SD(CO)	
JA(CO)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN	JAN. 17, 1977	speed variator	36D868812AA	
APPROVED	MAR. 18, 1977	ERIE, PA., USA	CONT ON SHEET FL	SH NO 2



NOTE:
1. ARROWS ON POTENTIOMETERS INDICATE THE WIPER DIRECTION AS THE SHAFT IS ROTATED CLOCKWISE

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

TELE ELEMENTARY DIAGRAM
REGULATOR CARD

FIRST MADE FOR STANDARD LINE
193X318X4501

36D86889I6AA

APPLIED PRACTICE

UNLESS OTHERWISE SPECIFIED SEE THE FOLLOWING
TOLERANCES ON DIMENSIONS
SURFACES
FRACTIONS
DECIMALS
ANGLES

DRAWN
CHK.
SPEED TRANSITION
EDITED PR.

APPROVALS
APRIL 1, 1972

RELEASED

REVISIONS

PRINTS TO
SERVICE
MFG.

36D86889I6AA

GENERAL ELECTRIC

36D868816AA

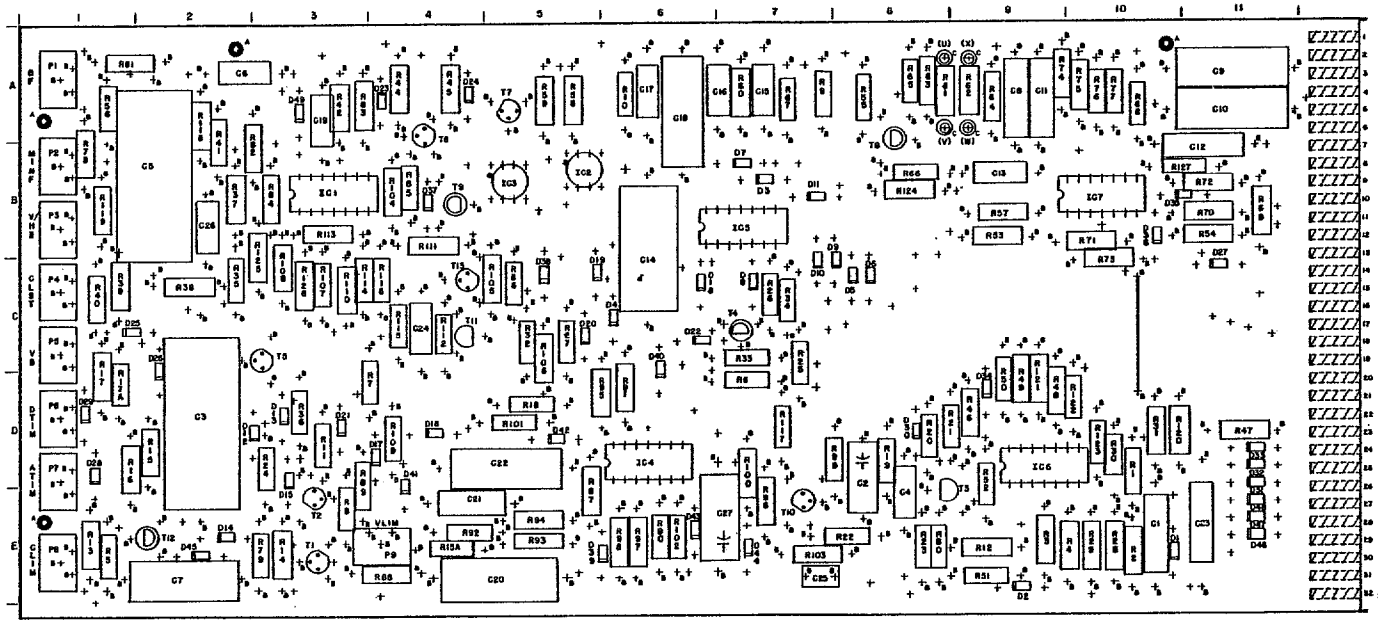
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCE TO MACHINED DIMENSIONS		
	✓	FRACTIONS	DECIMALS	ANGLES
		±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
REGULATOR CARD

FIRST MADE FOR AF-400

193X378AAG01



SEE NOTE 1&2

HOLE TABULATION

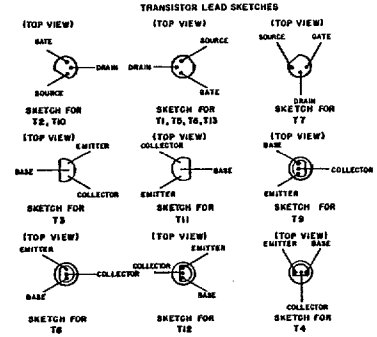
ALL HOLES ARE .032 EXCEPT THE HOLES TABULATED BELOW.

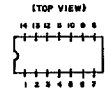
LOC.	DIA.	QTY.
A	.157	4
B	.040	374
C	.070	4

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, 11.50 +.000 −.016 X 6.150 +.000 −.008
4. ALL COMPONENTS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

TRANSISTOR LEAD SKETCHES

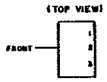




SKETCH FOR IC1, IC4, IC5, IC6, IC7

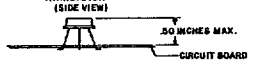


SKETCH FOR IC2, IC3



SKETCH FOR P1 THRU P8

COMPONENT HEIGHT SKETCH
TRANSISTOR
(SIDE VIEW)



1	0	2X	368605264AA005	368605265AA005
FRONT	BACK	SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.
SPACE UNIT REQUIREMENT				

REVISIONS

1	07-15-77	DWN. BY [illegible]	
2	12-29-77	CHK. BY [illegible]	

PRINTS TO

SB(BSM)	BJ(BW)
SM(BSM)	
SL(BSM)	
AW(BW)	
SD(CD)	
JA(CD)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

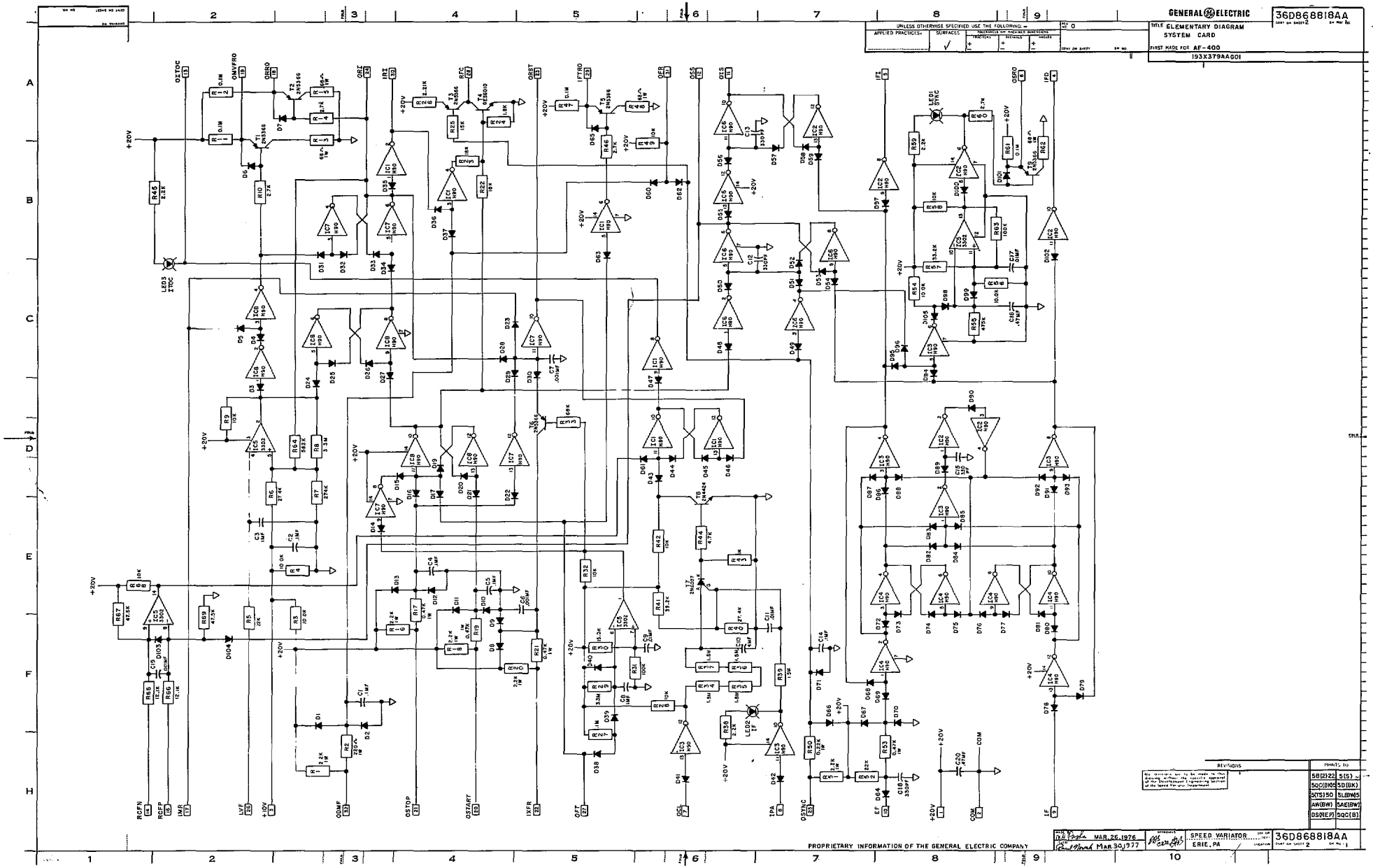
DWN. BY E. Gustin FEB. 21, 1977

APP'D. BY [illegible] DATE 11-22-77

speed variator

ERIE, PA., USA

36D868816AA


GENERAL ELECTRIC
36D868818AA
TITLE ELEMENTARY DIAGRAM
SYSTEM CARD
FIRST MADE FOR AF-400
193X379AAG01
SPEED VARIATOR
ERIE, PA
36D868818AA
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
MAR 26,1976
MAR 30,1977

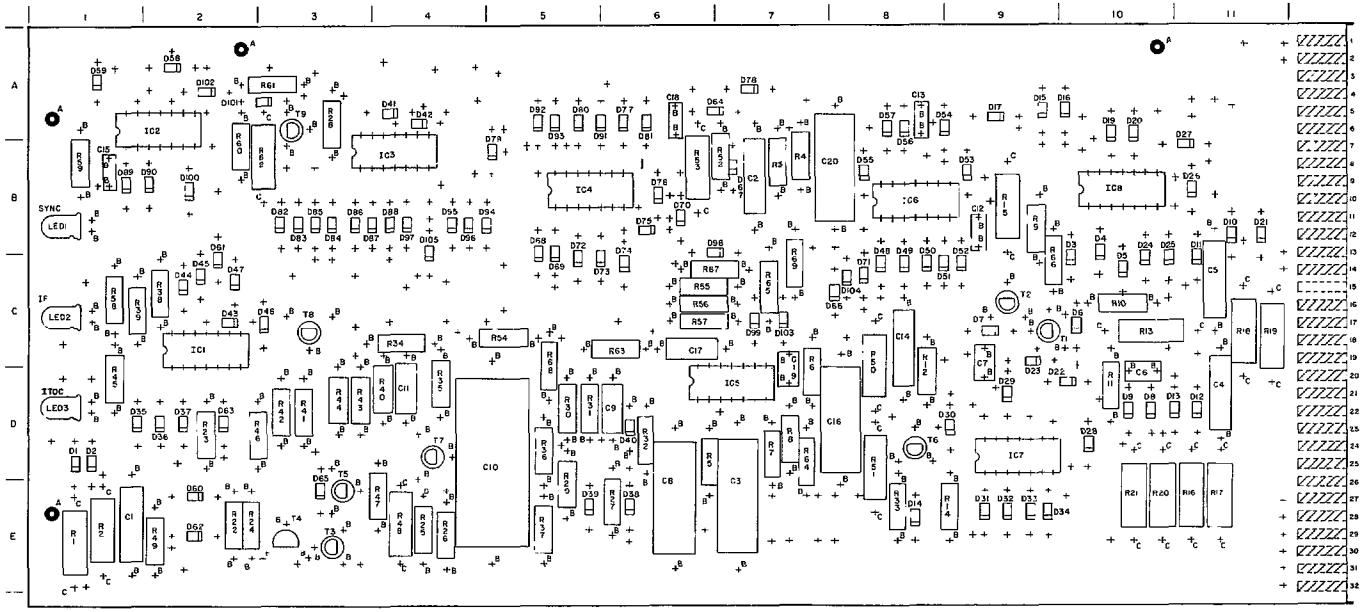
GENERAL ELECTRIC 36D868818AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES: | SURFACES: √ | REV. 0

TITLE PRINTED CIRCUIT DIAGRAM SYSTEM

FIRST MADE FOR AF-400

193X379AAG01



HOLE TABULATION

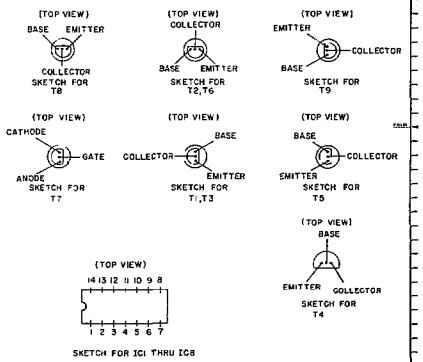
ALL HOLES ARE .032 EXCEPT THE HOLES TABULATED BELOW.

LOC.	DIA.	QTY.
A	.157	4
B	.040	183
C	.052	30

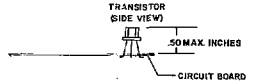
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, 11.50 +.000 −.015 × 5.130 +.003 −.005

TRANSISTOR LEAD SKETCHES



COMPONENT HEIGHT SKETCH



	FRONT	BACK
SPACE UNIT REQUIREMENT		
2X	36B605245AA005	36B605246AA005
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.

	SB(2)22 S(6)
	SOC(BK6) SO(BK)
	S(TS)50 SL(BW)5
	AW(BW) SAE(BW)
	DS(REF) SOC(B)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

Issued Jan 25,1977 | Revised Mar 30,1977 | speed varictor ERIE, PA., USA | 36D868818AA FL 2

GENERAL ELECTRIC

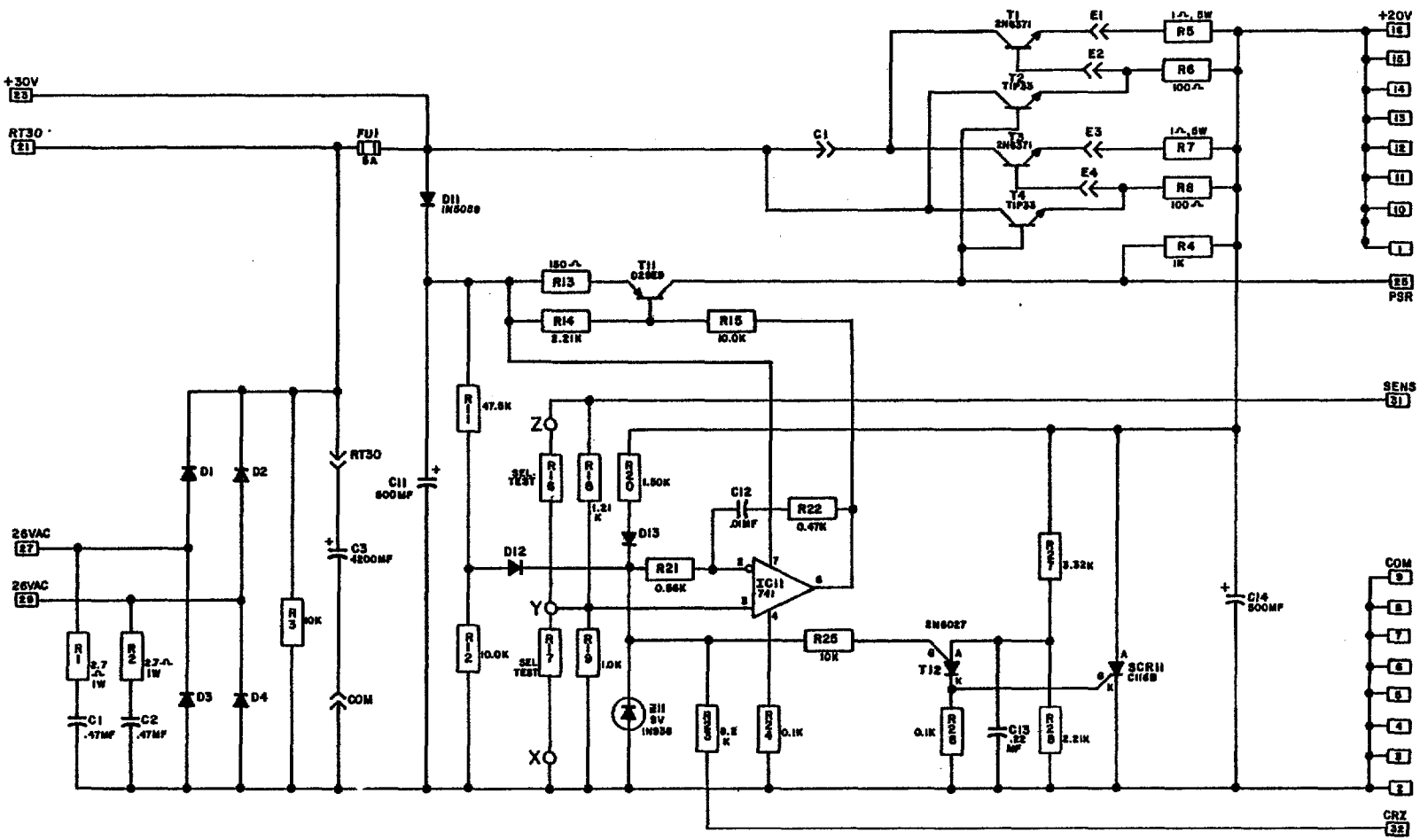
C764164AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
POWER SUPPLY

FIRST MADE FOR AF400
193X380AA001



FOLD

FOLD

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

REV NO. 0

MADE BY MD Pryle MAY 10, 1976
ISSUED Louis P Marsh FEB 1, 1977
APPROVALS MB EEZ BAS
SPEED VARIATOR — DIV OR DEPT
ERIE, PA — LOCATION
36C764164AA — CONT ON SHEET — SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

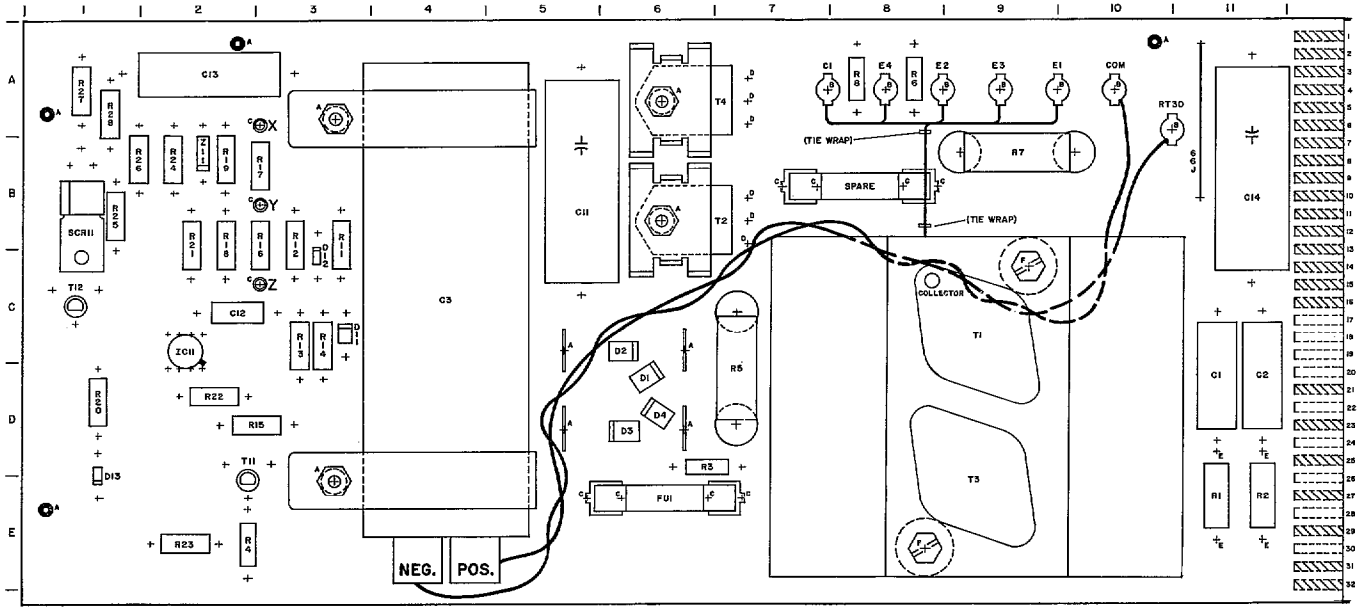
FF-500-F (5-75)
PRINTED IN U.S.A.

GENERAL ELECTRIC

36D868820AA

TITLE PRINTED CIRCUIT DIAGRAM
POWER SUPPLY
FIRST MADE FOR AF-400
193X380AA001

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES— | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±



SEE NOTE 1 & 2

HOLE TABULATION

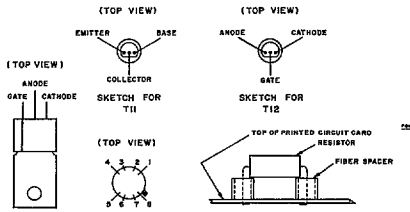
ALL HOLES ARE .040 EXCEPT THE HOLES TABULATED BELOW.

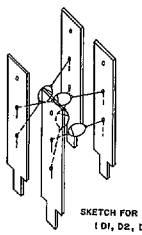
LOC.	DIA.	QTY.
A	.157	12
B	.101	7
C	.070	11
D	.062	5
E	.052	4
F	.189	2

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, 11.50 +.000 −.015 X 5.130 +.005 −.005

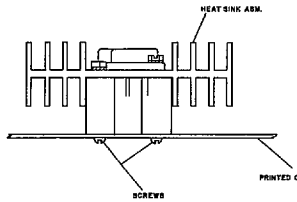
TRANSISTOR LEAD SKETCHES





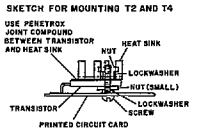
SKETCH FOR DIODE ASM. (D1, D2, D3, & D4)

COUPLE THE FOLLOWING	
WIRE NO.	CONNECTOR
T1E	E1
T1B	E2
T3E	E3
T3B	E4
C	C1
C3−	COM
C3+	RT30



SKETCH FOR MOUNTING TRANSISTOR HEAT SINK ASM.

SKETCH FOR MOUNTING T2 AND T4



2X	36B6O5266AA004	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.

REVISIONS

No revisions are to be made to this drawing without the specific approval of the responsible engineering section of the Speed Variator Department.

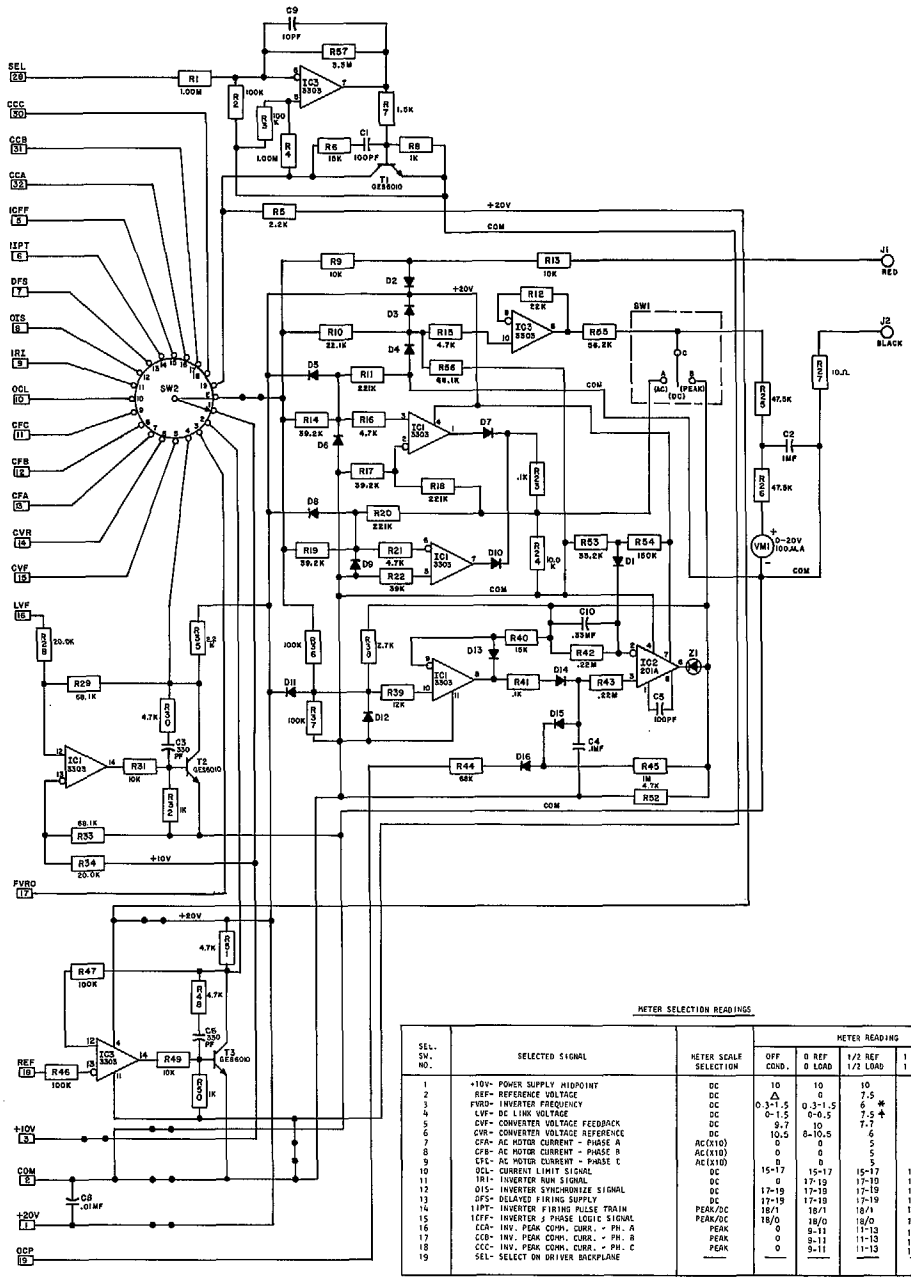
PRINTS TO

5A(2)22	5(S)
5OOB(W)5	5D(BK)
5(T)150	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

DRAWN BY C.R. Nelson JULY 13, 1976
CHECKED Ken Mont Feb 1, 1977
APPROVED [illegible]
speed variator ERIE, PA., USA

36D868820AA

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY



METER SELECTION READINGS

SEL. SW. NO.	SELECTED SIGNAL	METER SCALE SELECTION	METER READING				
			OFF COND.	0 REF 0 LOAD	1/2 REF 1/2 LOAD	1 REF 1 LOAD	SPECIAL CONDITION
1	+10V- POWER SUPPLY MIDPOINT	DC	10	10	10	10	10
2	REF- REFERENCE VOLTAGE	DC	△	0	7.5	15	——
3	FVRO- INVERTER FREQUENCY	DC	0-1.5	0-1.5	6 *	12 *	——
4	LVF- DC LINK VOLTAGE	DC	0-1.5	0-0.5	7.5 †	15 †	——
5	CVF- CONVERTER VOLTAGE FEEDBACK	DC	9.7	10	7-7	5-5	——
6	CVR- CONVERTER VOLTAGE REFERENCE	DC	10.5	8-10.5	6	3-5	——
7	CFA- AC MOTOR CURRENT - PHASE A	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
8	CFB- AC MOTOR CURRENT - PHASE B	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
9	CFC- AC MOTOR CURRENT - PHASE C	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
10	OCL- CURRENT LIMIT SIGNAL	DC	15-17	15-17	15-17	15-17	4-8 AT CURR. LIM.
11	IRI- INVERTER RUN SIGNAL	DC	0	17-19	17-19	17-19	——
12	OIS- INVERTER SYNCHRONIZE SIGNAL	DC	17-19	17-19	17-19	17-19	0 AT SYNCH.
13	DFS- DELAYED FIRING SUPPLY	DC	17-19	17-19	17-19	17-19	——
14	IIPT- INVERTER FIRING PULSE TRAIN	PEAK/DC	18/1	18/1	18/1	18/1	——
15	ICFF- INVERTER 3 PHASE LOGIC SIGNAL	PEAK/DC	18/0	18/0	18/0	18/0	——
16	CCA- INV. PEAK COMM. CURR. - PH. A	PEAK	0	9-11	11-13	12-14	——
17	CCB- INV. PEAK COMM. CURR. - PH. B	PEAK	0	9-11	11-13	12-14	——
18	CCC- INV. PEAK COMM. CURR. - PH. C	PEAK	0	9-11	11-13	12-14	——
19	SEL- SELECT ON DRIVER BACKPLANE	——	——	——	——	——	——

* READINGS SHOWN ARE FOR 60HZ BASE FREQUENCY SETTING. OTHER READINGS WILL APPLY FOR OTHER BASE FREQUENCIES. ACTUAL FREQUENCIES IN HERTZ IS 5 TIMES METER READING FOR THE 75HZ MAX. FREQUENCY SELECTION AND 10 TIMES THE METER READING FOR THE 150HZ MAX. FREQUENCY SELECTION.

† ACTUAL DC LINK VOLTAGE IS 40 TIMES THE METER READING FOR 460 VOLT DRIVES AND 20 TIMES THE METER READING FOR 230 VOLT DRIVES.

△ DEPENDENT ON REFERENCE SETTING.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC
ERIE, PA
SPEED VARIATOR
TITLE ELEMENTARY CARD DIAGRAM METER CARD
FIRST MADE FOR STANDARD LINE
193X391AA601
360D868822AA
SHT. 1 OF 2

GENERAL ELECTRIC 36D868822AA

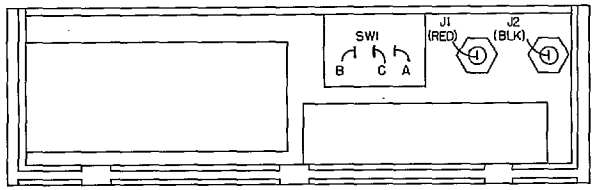
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES: — SURFACES: ✓ — TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± , DECIMALS ± , ANGLES ± — MATL Ø 2 — CONT ON SHEET — SH NO.

TITLE PRINTED CIRCUIT DIAGRAM METER CARD — CONT ON SHEET FL — SH NO. 2

FIRST MADE FOR AF-400

193X381AAG01

CARD COVER
BACK VIEW SHOWING LOCATION OF COMPONENTS & WIRING INFORMATION



HOLE TABULATION

ALL HOLES ARE .040 EXCEPT THE HOLES TABULATED BELOW.

LOC.	DIA.	QTY.
A	.157	8
B	.062	27
C	.032	68

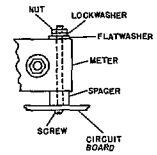
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $11.50^{+.000}_{-.015} \times 5.130^{+.005}_{-.005}$
4. ALL WIRING IS TERMINATED TO THE PRINTED CIRCUIT ON ONE END. THE OTHER WIRE TERMINATION IS DESIGNATED BY COMPONENT AND TERMINAL NEAR THE PRINTED CIRCUIT BOARD TERMINATION.

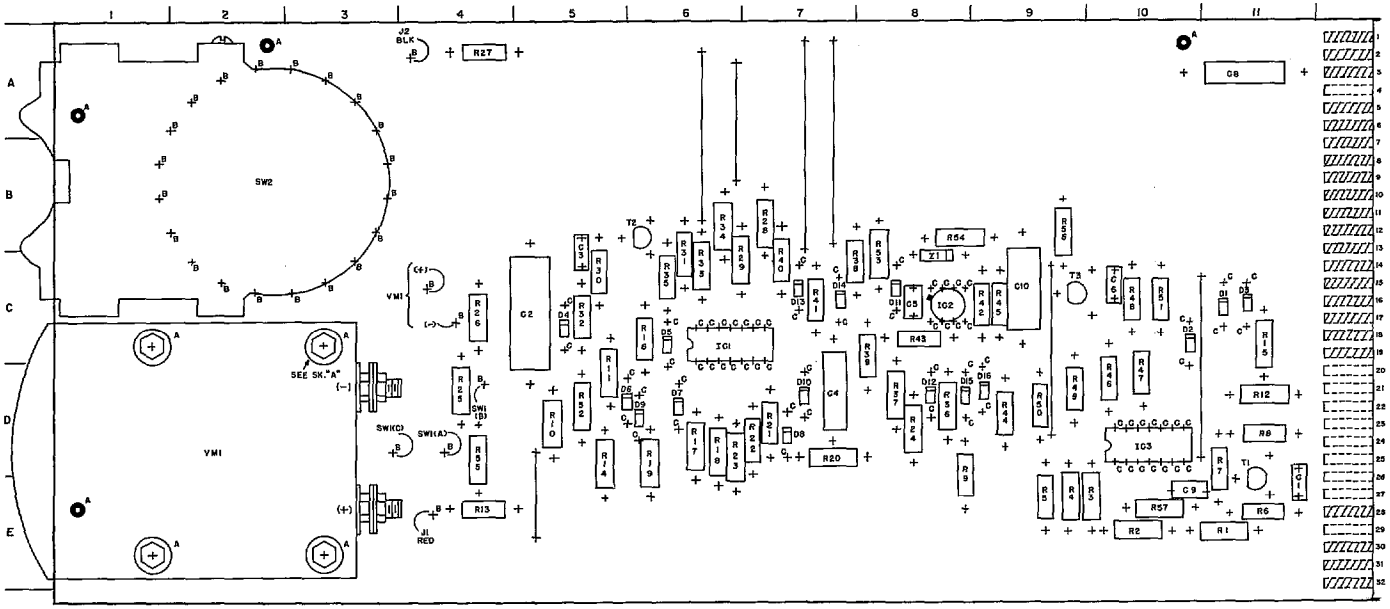
TRANSISTOR LEAD SKETCHES



SKETCH FOR T1,T2,T3



SKETCH "A"
MOUNTING OF METER TO PRINTED CIRCUIT BOARD



PRINTED CIRCUIT BOARD
FRONT VIEW SHOWING LOCATION OF COMPONENTS & WIRING INFORMATION

SPARE UNIT REQUIREMENT	
FRONT	BACK
2	—
2X 36B605263AA004	—
SCALE — REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.

REVISIONS

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)1	5AE(BW)
D5(REP)	5QC(B)

Sal Acquisto July 13 1977 — APPROVED — gps — speed variator — ERIE, PA., USA — 36D868822AA — CONT ON SHEET FL — SH NO. 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764166AA

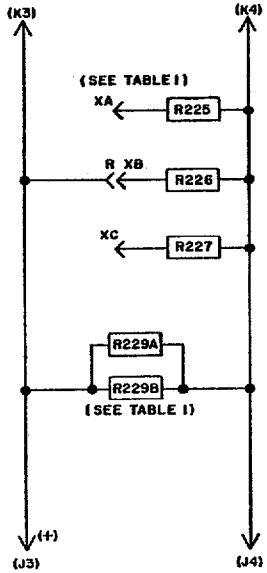
SCHEMATIC DIAGRAM

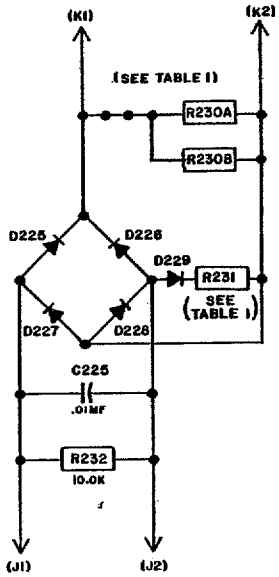
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES: 718A515 | SURFACES: ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
CURRENT FEEDBACK

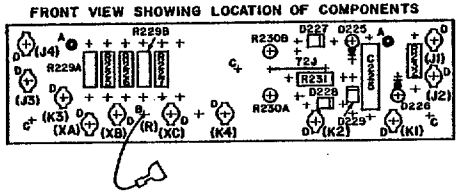
FIRST MADE FOR AF400

193X382AA601—G05





FRONT VIEW SHOWING LOCATION OF COMPONENTS



HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.062	1
C	.187	3
D	.101	11

NOTE: 1. CARD SIZE, 4.900 $^{+.000}_{-.015}$ X 1.300 $^{+.000}_{-.015}$

TABLE 1

GROUP NUMBER	RESISTOR NOMENCLATURE AND VALUE IN OHMS							
	XA R225	XB R226	XC R227	R229A	R229B	R230A	R230B	R231
G01	1210	121	68.1	68.1	182	100	—	562
G02	562	82.5	33.2	27.4	274	100	100	274
G03	392	47.5	18.2	12.1	475	33.2	100	150
G04	681	56.2	27.4	15.0	562	22.1	68.1	100
G05	68.1	27.4	18.2	12.1	150	22.1	27.4	68.1

TABLE 2

GROUP NUMBER	JUMPER SELECTION	RESISTANCE K3 TO K4 OHMS	C.T. TURNS RATIO	NOMINAL INV. RMS AMPS
G01	XA	47.6	2000/1	42
↓	XB	35.2	↓	56
	XC	28.7		70
G02	XA	23.8	2000/1	84
↓	XB	19.1	↓	105
	XC	14.2		140
G03	XA	11.45	2000/1	175
↓	XB	9.45	↓	210
	XC	7.16		280
G04	XA	14.3	4000/1	280
↓	XB	11.6	↓	350
	XC	9.53		420
G05	XA	9.62	4000/1	420
↓	XB	7.95	↓	500
	XC	6.93		580

$$\text{VOLTS (K3–K4)} = \frac{\text{INV. RMS AMPS}}{\text{C.T. TURNS RATIO}} \times \text{OHMS (K3–K4)}$$

= 1.0 VOLT RMS NOMINAL

TABLE 3

GROUP NUMBER	RESISTANCE K1 TO K2 OHMS	C.T. TURNS RATIO	COMM. PEAK AMPS
G01	100	2000/1	250
G02	50	2000/1	500
G03	24.9	2000/1	1000
G04	16.7	2000/1	1500
G05	12.2	2000/1	2050

$$\text{VOLTS (K1–K2)} = \frac{\text{COMM. PEAK AMPS}}{\text{C.T. TURNS RATIO}} \times \text{OHMS (K1–K2)}$$

= 12.5 VOLTS PEAK NOMINAL FOR THE POSITIVE COMMUTATION.

NOTE 2: FOR THE NEGATIVE COMMUTATION, A MULTIPLIER OF 0.65 IS APPLIED TO THE ABOVE FORMULA DUE TO THE LOADING EFFECT OF R231.

REV NO. Ø 1

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	36B605257AA004		

REVISIONS

1	JUNE 1, 1977 R.R. Baker	CHG. PER SV-872 @ SC	CK'D

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

5QC(BK)5	5(S)
5B(2)22	5D(BK)
5L(BW)5	
5(TS)50	
JA(CD)	
AW(BW)	5(QC)8

MADE BY R.R. Baker AUG. 17, 1976
ISSUED R.R. Baker JAN. 19, 1977
APPROVALS cek jar dhs
speed variator DIV OR DEPT
ERIE, PA. LOCATION
36C764166AA CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC | 36C764172AA

SCHEMATIC DIAGRAM

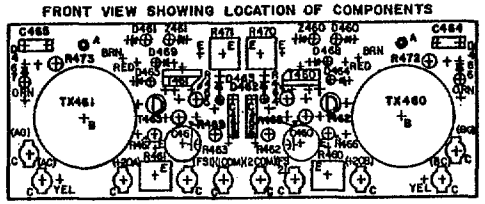
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
PULSE TRANSFORMER

FIRST MADE FOR AF-400

193X383AAG01

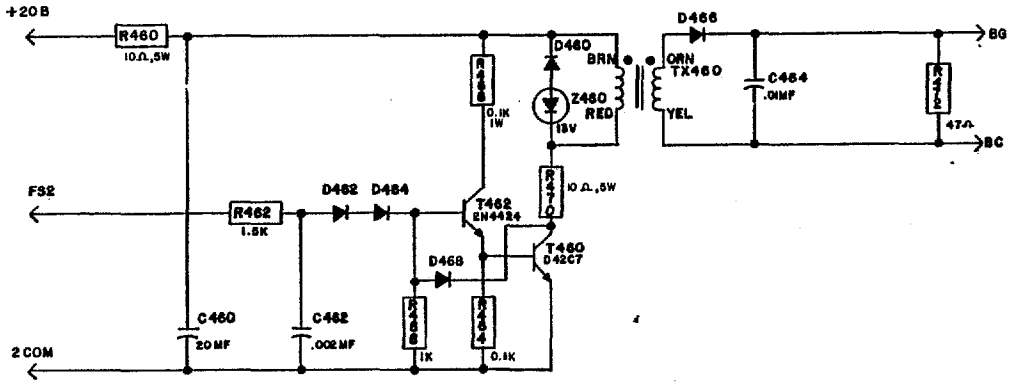
FRONT VIEW SHOWING LOCATION OF COMPONENTS

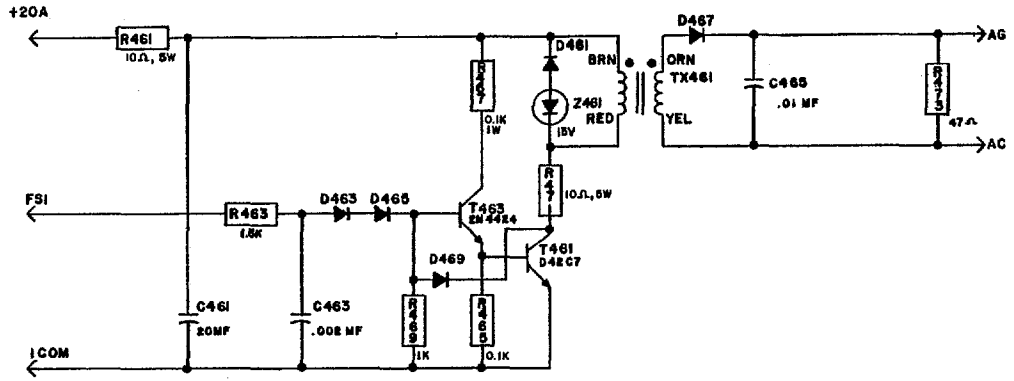


HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

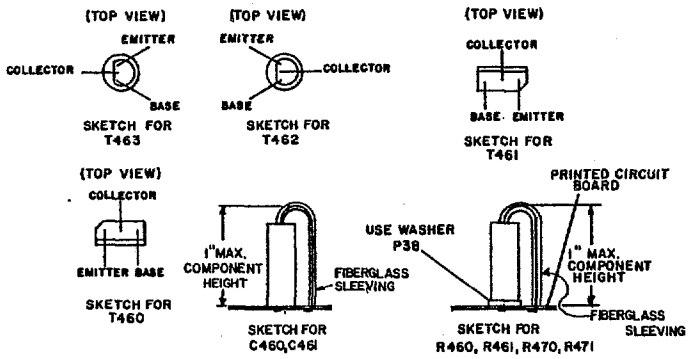
LOC.	DIA.	QUAN.
A	.157	2
B	.187	2
C	.101	10
E	.052	8





NOTE: 1. CARD SIZE 5.130 $^{+.000}_{-.015}$ X 1.90 ± .020

TRANSISTOR LEAD SKETCHES



SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	36B605270AA004	———	———

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

J	VCode per SV572**DD		

PRINTS TO

5QC(BK)5	5B(S)
5B(2)22	5D(BK)
5(TS)50	5(BW)5
DS(REP)	5AE(BW)
	AW(BW)
	5(QC)6

MADE BY	DATE	APPROVALS		
M.B. Taylor	JULY 28,1976		SPEED VARIATOR	36C764172AA
TRACED			ERIE, PA	
Paul Novak	APR. 1, 1977	ese	LOCATION	CONT ON SHEET / SH NO.

GENERAL ELECTRIC

36D868888AA

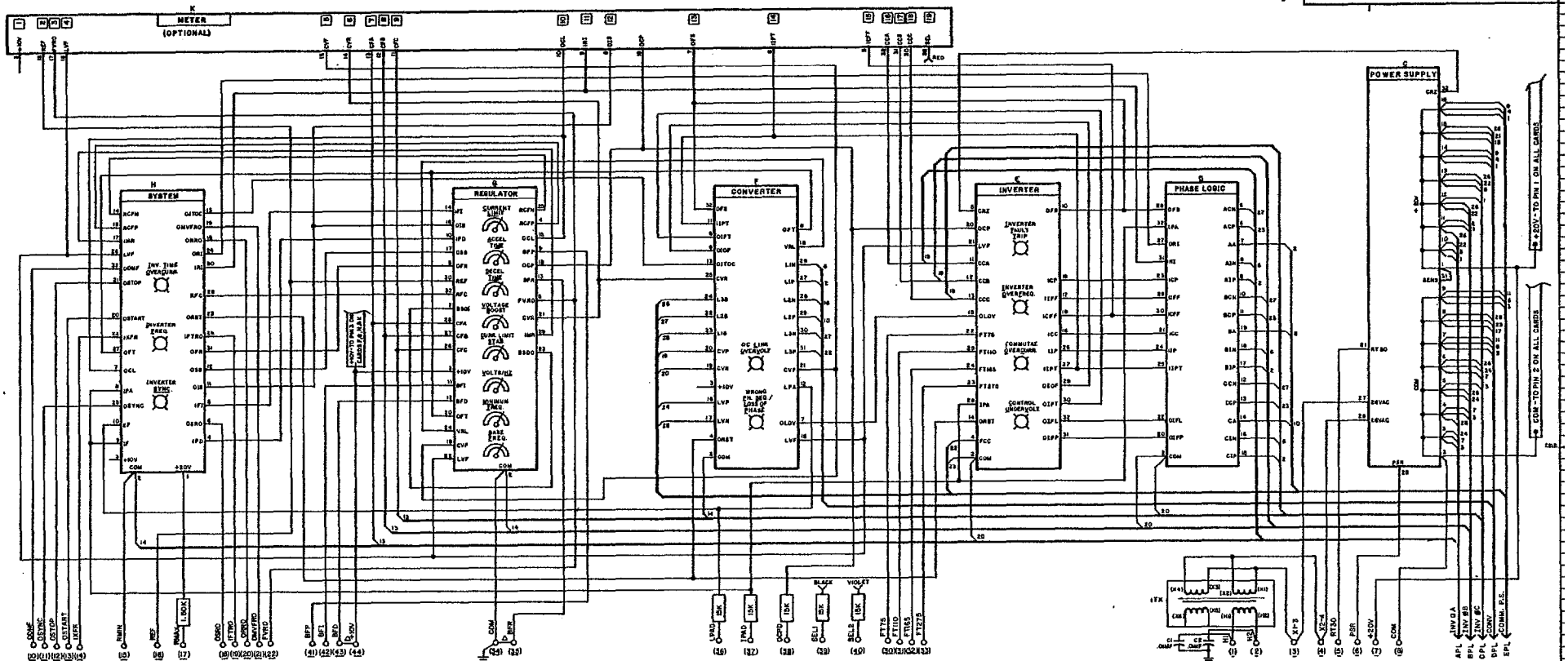
TITLE ELEMENTARY DIAGRAM

DRIVER

FIRST MADE FOR AF-400

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:

APPLIED PRACTICES:	SURFACES	DIMENSIONS	DECIMALS	ANGLES
	√	±	±	±



DRIVER SYMBOLS

THE LETTERS ABOVE THE CARD BLOCKS REFER TO THE LOCATION OF THE CARD IN THE DRIVER RACK

CARD MOUNTED INDICATING LIGHT

CARD MOUNTED ADJUSTMENT POTENTIOMETER

NUMBERS IN SQUARES IN METER CARD BLOCK REFER TO SELECTOR SWITCH POSITIONS FOR READING ADJACENT SIGNAL INPUTS

CONTROL POWER BUS BAR FOR CARD BACK PLANE INTERCONNECTION

THE HEAVY LINE REPRESENTS THE WIRE BUNDLES LEADING TO THE CONNECTOR PLUG DENOTED BY THE ARROW AND LOCATED AT THE BOTTOM OF THE CARD RACK. THE CONNECTOR PLUG PIN NUMBER IS SHOWN WHERE EACH WIRE LEAVES THE WIRE BUNDLE.

THE NUMBER SHOWN ABOVE EACH WIRE, AS IT LEAVES THE CARD BLOCK, IS THE CARD TAB NUMBER AND ALSO THE CARD RECEPTACLE NUMBER.

INDICATES TERMINAL BOARD JUMPER

INDICATES TERMINATION OF WIRE AT TERMINAL BOARDS LOCATED AT BOTTOM OF CARD RACK.

INDICATES RESISTOR IN WIRE LEADING FROM CARD RECEPTACLE TO TERMINAL BOARD

DRIVER NOMENCLATURE

THE ABBREVIATED NOMENCLATURE USED ON THE DRIVER IS LISTED BELOW. A "1" OR "0" PRECEDING THE ABBREVIATION INDICATES RESPECTIVELY THE HIGH (+20 VOLTS) OR LOW (0 VOLTS) LOGIC STATE WHICH PRODUCES OR INDICATES THE SPECIFIED OPERATION OR SIGNAL.

(A,B) CA	- PHASE (A,B) C AUXILIARY FIRING SIGNAL
(A,B) CCH	- PHASE (A,B) C COMMUTATION FIRING SIGNAL - NEG. BUS
(A,B) CCP	- PHASE (A,B) C COMMUTATION FIRING SIGNAL - POS. BUS
(A,B) CIN	- PHASE (A,B) C INVERTER FIRING SIGNAL - NEG. BUS
(A,B) CIP	- PHASE (A,B) C INVERTER FIRING SIGNAL - POS. BUS
BFD(I)	- BASE FREQUENCY DECREASE (INCREASE) SIGNAL INPUT
BFP	- BASE FREQUENCY EXTERNAL POTENTIOMETER SUPPLY
CCA (B,C)	- COMMUTATION CURRENT SIGNAL - PHASE A (B,C)
CCC	- COMMUTATION CURRENT LOGIC SIGNAL
CFA (B,C)	- MOTOR CURRENT FEEDBACK SIGNAL - PHASE A (B,C)
CFF	- COMMUTATION FLIP - FLOP TRIGGER
CCL	- CURRENT LIMIT LOGIC SIGNAL
BFR	- BASE FREQUENCY RANGE SELECTION
COC	- COMMUTATION OVERCURRENT TRIP SIGNAL
(0) ICP	- COMMUTATION PULSE
OCPD	- COMMUTATION PULSE DIAGNOSTIC TRIGGER
CRZ	- CONTROL REFERENCE ZENER VOLTAGE
CUV	- CONTROL UNDERVOLTAGE TRIP SIGNAL
CVF	- CONVERTER VOLTAGE FEEDBACK
CVN(P)	- CONVERTER OUTPUT VOLTAGE - NEG. (POS.) BUS
CVR	- CONVERTER VOLTAGE REFERENCE
DFS	- DELAYED FIRING SIGNAL SUPPLY
DDRF	- DECELERATE TO MINIMUM FREQUENCY SIGNAL
EF	- EXTERNAL FREQUENCY SIGNAL
FCC	- FILTER CAPACITOR CURRENT SIGNAL
IFD (I)	- FREQUENCY DECREASE (INCREASE) - SYNCHRONIZED
IFR	- FAST ACCELERATION - DECELERATION RATE SIGNAL
FF75(110,165,275)	- OVERFREQUENCY TRIP SELECTION - 75 HZ (110,165,275)
IFT	- FAULT TRIP SIGNAL
IFTRO	- FAULT TRIP READOUT SIGNAL
FVRO	- FREQUENCY VOLTAGE READOUT SIGNAL
IF	- INVERTER FREQUENCY SIGNAL
IFFF	- INVERTER FLIP-FLOP TRIGGER
IFFL	- INVERTER FAULT FIRING LOCKOUT
IFFP	- INVERTER FAULT FIRING PULSE
(0)IFT	- INVERTER FAULT TRIP SIGNAL
(0)IOF	- INVERTER OVERFREQUENCY TRIP SIGNAL
IPA	- INVERTER PHASE A LOGIC SIGNAL
IPAD	- INVERTER PHASE A DIAGNOSTIC SIGNAL
IIP	- INITIAL FIRING PULSE
IIPT	- INVERTER FIRING PULSE TRAIN
ISS	- INVERTER SYNCHRONIZING LOGIC SIGNAL
L1(2,3)N	- PHASE 1 (2,3) CONVERTER FIRING SIGNAL (NEG. BUS)
L1(2,3)P	- PHASE 1 (2,3) CONVERTER FIRING SIGNAL (POS. BUS)
(0)LOV	- DC LINK OVERVOLTAGE TRIP SIGNAL
LPA	- A.C. LINE PHASE A LOGIC SIGNAL
LPAD	- A.C. LINE PHASE A DIAGNOSTIC SIGNAL
L1(2,3)S	- A.C. LINE PHASE 1(2,3) SYNCHRONIZING SIGNAL
LVF	- D.C. LINK VOLTAGE FEEDBACK
LVN(P)	- D.C. LINK VOLTAGE - NEG. (POS.) BUS
MMR	- MINIMUM REFERENCE SIGNAL
OMVFRO	- MINIMUM VOLTAGE AND FREQUENCY READOUT
PS/LOP	- WRONG PHASE SEQUENCE/LOSS OF PHASE TRIP SIGNAL
PSR	- POWER SUPPLY REGULATION SIGNAL
REF	- REFERENCE INPUT SIGNAL
RFC	- REFERENCE INPUT CLAMP
(0)IRE	- RUN INVERTER LOGIC SIGNAL
RMAX	- REFERENCE POTENTIOMETER CONNECTION - MAXIMUM
RMIN	- REFERENCE POTENTIOMETER CONNECTION - MINIMUM
ORRO	- INVERTER RUNNING READOUT SIGNAL
OSST	- FAULT RESET SIGNAL
RT30	- CONTROL POWER AIDE - THROUGH +30 VOLT INPUT
SEL(1,2)	- SIGNAL SELECTOR FOR DIAGNOSTIC
SENS	- POWER SUPPLY VOLTAGE SENSING
OSS	- START SYNCHRONIZING INVERTER LOGIC SIGNAL
OSRO	- INVERTER SYNCHRONIZED READOUT SIGNAL
SSD1(0)	- INVERTER STABILITY/SLOWDOWN INPUT (OUTPUT)SIGNAL)
OSTART	- DRIVE START INPUT SIGNAL
OSTOP	- DRIVE STOP INPUT SIGNAL
OSYNC	- SYNCHRONIZE INVERTER INPUT SIGNAL
SYNC	- INVERTER SYNCHRONIZED SIGNAL
VRL	- CONVERTER VOLTAGE REFERENCE LIMIT SIGNAL
IXFR	- EXTERNAL FAULT RESET INPUT SIGNAL
+20V	- +20 VOLT REGULATED POWER SUPPLY
+10V	- +10 VOLT REGULATED POWER SUPPLY
COM	- COMMON
(0)ITOC	- INVERSE TIME OVERCURRENT TRIP SIGNAL
RCFN(P)	- RECTIFIED MOTOR CURRENT FEEDBACK - NEG. (POS.)

NOTE: SEE DRIVER NOTES ON SH. 2 THRU SH. 8 FOR DETAILED INFORMATION.

REVISIONS			PRINTS TO
1			5(a) 5A(b)
2			5B(a) 5B(b)
3			5C(a) 5C(b) 5S
			5T5(a)

DRAWN	CHECKED	APPROVED	SPEED VARIATOR	36D868888AA
MAR 3, 1977			ERIE, PA	SHT 2

PP-852-P (8-78) PRINTED IN U.S.A.

IN THESE NOTES, A "HIGH" LOGIC STATE REFERS TO +18 TO +20 VOLTS AND A "LOW" LOGIC STATE REFERS TO 0 TO +1 VOLT.

A. DRIVER INPUTS

1. ODMF - IF TB10 IS TAKEN LOW WHEN THE INVERTER IS RUNNING, THE INVERTER WILL DECELERATE TO ITS MINIMUM FREQUENCY AT A FAST RATE, AND WILL REMAIN THERE AS LONG AS THE ODMF INPUT IS IN THE LOW STATE. WHEN ODMF GOES HIGH, THE INVERTER WILL RETURN TO THE REFERENCE LEVEL FREQUENCY AT A NORMAL ACCELERATION RATE.

2. OSYNC - IF TB11 IS TAKEN LOW WHEN THE INVERTER IS RUNNING, THE INVERTER WILL ACCELERATE AT A FAST RATE TO THE A.C. SUPPLY LINE FREQUENCY, AND SYNCHRONIZE TO THAT FREQUENCY. THE INVERTER WILL REMAIN SYNCHRONIZED TO THE LINE FREQUENCY AS LONG AS THE OSYNC INPUT REMAINS IN THE LOW STATE, AND WILL RETURN TO THE REFERENCE LEVEL FREQUENCY AT A NORMAL RATE WHEN OSYNC GOES HIGH.

3. OSTOP - IF TB12 IS TAKEN LOW WHEN THE INVERTER IS RUNNING, THE INVERTER WILL DECELERATE TO ITS MINIMUM FREQUENCY AT A NORMAL RATE AND THEN STOP OPERATING. THE INVERTER WILL REMAIN OFF AS LONG AS THE OSTOP INPUT IS LOW IRRESPECTIVE OF ANY OTHER INPUT SIGNALS. THE ABOVE STOPPING PROCEDURE WILL ALSO OCCUR WHEN OSTOP MOMENTARILY GOES LOW IF THE OSTART INPUT IS IN THE HIGH STATE.

4. OSTART - IF TB13 IS MOMENTARILY TAKEN LOW WHEN THE INVERTER IS OFF AND THE OSTOP (TB12) INPUT IS IN THE HIGH STATE, THE INVERTER WILL START OPERATING AT ITS MINIMUM FREQUENCY, AND WILL ACCELERATE AT A NORMAL RATE TO THE REFERENCE LEVEL FREQUENCY.

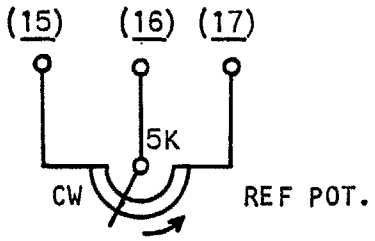
5. IXFR - IF TB14 IS HELD LOW, A FAULT SHUTDOWN CANNOT BE RESET UNTIL THE EXTERNAL FAULT RESET (TB14) INPUT IS MOMENTARILY TAKEN HIGH. WHEN TB14 IS TAKEN HIGH, OR IS LEFT UNCONNECTED, A FAULT SHUTDOWN WILL BE RESET WHEN THE OSTOP INPUT IS LOW AND THE LOGIC SIGNALS ORI AND 1MR ARE IN THEIR HIGH STATES.

6. REF, RMIN, RMAX - THE REFERENCE VOLTAGE LEVEL AT TB16 DETERMINES THE INVERTER FREQUENCY AND VOLTAGE WHEN THE INVERTER IS OPERATING NORMALLY (ODMF AND OSYNC INPUTS BOTH HIGH). IF A REFERENCE POTENTIOMETER IS USED, IT SHOULD BE 5000 OHMS IN VALUE AND HAVE ITS MINIMUM END CONNECTED TO RMIN (TB15) AT ZERO VOLTS, AND ITS MAXIMUM END CONNECTED TO RMAX (TB17), AS SHOWN IN THE SKETCH BELOW, SO THAT THE MAXIMUM REFERENCE VOLTAGE IS +15 VOLTS. IF SOME OTHER OHMIC VALUE POTENTIOMETER IS USED, ITS MAXIMUM END SHOULD BE CONNECTED IN SERIES WITH A SUITABLE VALUE RESISTOR TO +20V (TB7) SO THAT THE MAXIMUM REFERENCE VOLTAGE IS +15 VOLTS. IF SOME OTHER VOLTAGE SOURCE IS USED AS A REFERENCE, IT SHOULD BE CONNECTED BEWEEN TB16 AND TB15 (TB16 POSITIVE) AND SHOULD BE 15 VOLTS FOR MAXIMUM REFERENCE.

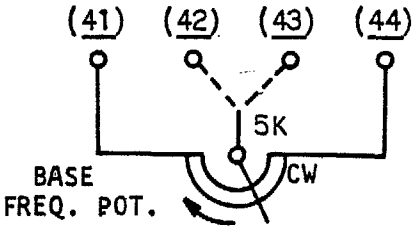
JA(CD)
5TS(50)
5QC(BK)5
5QC(B)
5D(BK)
5B(3)23
5(S)
PRINTS TO

drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

6. (CONT.)



7. BFP, BFI, BFD - IF AN EXTERNAL BASE FREQUENCY ADJUSTING POTENTIOMETER IS USED, IT SHOULD BE 5000 OHMS IN VALUE AND SHOULD BE CONNECTED FROM BFP (TB41) TO +10V (TB44) AS SHOWN IN THE SKETCH BELOW. ITS WIPER ARM SHOULD BE CONNECTED EITHER TO BFI (TB42) TO INCREASE THE THE BASE FREQUENCY OR TO BFD (TB43) TO DECREASE THE BASE FREQUENCY. THIS WILL PROVIDE EXTERNAL ADJUSTMENT OF THE BASE FREQUENCY BY AS MUCH AS 50% ABOVE OR BELOW THE INTERNALLY SET BASE FREQUENCY, WITHIN THE MAXIMUM FREQUENCY RATING OF THE DRIVE. THE JUMPER BETWEEN TB43 AND TB44 SHOULD BE REMOVED ONLY IF THE EXTERNAL POTENTIOMETER WIPER IS CONNECTED TO TB43.



B. DRIVER LINE SYNCH AND VOLTAGE FEEDBACKS

1. L1S, L2S, L3S - TO PROVIDE FOR PHASE CONTROL LINE SYNCHRONIZATION OF THE CONVERTER; DRIVER PLUG EPL IS CONNECTED TO L1, L2 AND L3 PHASES RESPECTIVELY OF THE A.C. SUPPLY THROUGH SUITABLE SERIES RESISTORS. THESE RESISTORS ARE PLACED IN THE CONNECTING WIRE HARNESS AND ARE 1.5 MEGOHMS FOR A 230 VOLT SUPPLY AND 3.0 MEGOHMS FOR A 460 VOLT SUPPLY.

2. CVP, CVN - TO PROVIDE CONVERTER VOLTAGE FEEDBACK, DRIVER PLUG EPL IS CONNECTED TO THE CONVERTER MODULE OUTPUT, WITH CVP POSITIVE, THROUGH SUITABLE SERIES RESISTORS. THESE RESISTORS ARE PLACED IN THE CONNECTING WIRE HARNESS AND ARE 1.5 MEGOHMS FOR A 230 VOLT CONVERTER AND 3.0 MEGOHMS FOR A 460 VOLT CONVERTER.

3. LVP, LVN - TO PROVIDE D.C. LINK VOLTAGE FEEDBACK, DRIVER PLUG EPL IS CONNECTED TO THE INVERTER D.C. LINK, WITH LVP POSITIVE, THROUGH SUITABLE SERIES RESISTORS. THESE RESISTORS ARE PLACED IN THE

REVISIONS

1	MARCH 23, 1977	PER ECR V1V0anaaa6V-872 #1C REVISED WITH GENERAL CHGS. RECORD COPY FILED

JA(CD)
5TS(50)
5QC(BK)5
5QC(8)
5D(8K)
58(3)23
5(S)
PRINTS TO

Property of the Development Engineering Section of the Speed Variator Department.

CONNECTING WIRE HARNESS AND ARE 1.5 MEGOHMS FOR A 300 VOLT D.C. LINK AND 3.0 MEGOHMS FOR A 600 VOLT D.C. LINK.

C. DRIVER JUMPER SELECTIONS

1. BFR - TB35 IS THE MEANS TO CHANGE THE INVERTER BASE FREQUENCY RANGE AS FOLLOWS:
   37.5 TO 75 HZ - JUMPER TB35 TO TB34 (COM)
   75 TO 150 HZ - LEAVE TB35 OPEN.

   BASE FREQUENCY IS THE FREQUENCY AT WHICH THE INVERTER REACHES FULL VOLTAGE AND IS ADJUSTED BY THE BASE FREQ. POTENTIOMETER WITHIN EITHER OF THE ABOVE TWO RANGES. ALTHOUGH THIS DIAGRAM SHOWS BFR JUMPERED TO COM., CONSULT THE DRIVE ELEMENTARY DIAGRAM TO DETERMINE IF THIS JUMPER SHOULD BE CONNECTED OR NOT.
   CAUTION: IMPROPER JUMPER CONNECTION MAY RESULT IN DRIVE MALFUNCTION OR DAMAGE.

2. FT75, FT110, FT165, FT275 - TB30, TB31, TB32 AND TB33 ARE THE MEANS TO SELECT AN UPPER FREQUENCY LIMIT AT WHICH THE DRIVE WILL TRIP AND SHUT DOWN, TO PREVENT MOTOR OVERSPEED. THE FREQUENCY TRIP LEVELS ARE SELECTED BY JUMPERS AS FOLLOWS:
   75 HZ TRIP - NO JUMPER
   110 HZ TRIP - JUMPER TB30 TO TB31
   165 HZ TRIP - JUMPER TB30 TO TB32
   275 HZ TRIP - JUMPER TB30 TO TB33

   CONSULT THE DRIVE ELEMENTARY DIAGRAM FOR THE PROPER JUMPER PLACEMENT.
   WARNING: IMPROPER JUMPER PLACEMENT MAY RESULT IN DRIVE MALFUNCTION OR PRESENT AN EQUIPMENT OR PERSONNEL HAZARD DUE TO MOTOR OVERSPEED.

D. CARD JUMPER SELECTIONS

1. 460V JUMPER ON INVERTER CARD (193X376AA G01) - THIS JUMPER SHOULD BE PRESENT ON ALL 460 VOLT OUTPUT INVERTER DRIVES FOR PROPER OPERATION. THIS JUMPER SHOULD BE REMOVED ON ALL 230 VOLT OUTPUT INVERTER DRIVES FOR PROPER OPERATION.
   CAUTION: INCORRECT JUMPER CONNECTION WILL RESULT IN MALFUNCTION AND POSSIBLE DAMAGE TO THE INVERTER.

2. 60 HZ JUMPER ON CONVERTER CARD (193X377AA G01) - THIS JUMPER SHOULD BE PRESENT ON ALL DRIVES SUPPLIED FROM 60HZ A.C. POWER FOR PROPER SYNCHRONIZING OF THE INVERTER TO THE A.C. SUPPLY (WHEN THIS IS DONE). FOR DRIVES SUPPLIED FROM 50 HZ A.C. POWER, THIS JUMPER SHOULD BE REMOVED.

REVISIONS

D	JULY 13, 1977 M. Cotty	SV-872 #11E DELETED PART of NOTE 1+2 clh

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Varrator Department.

JA(CD)
5TS(50)
5QC(BK)5
5QC(B)
5D(BK)
5B(3)23
5(S)
PRINTS TO

REVISIONS

D	JULY 13, 1977 M. Seth	SV-872 #11E DELETED PART of NOTE 5 CDL

E. ADJUSTMENTS

ALL DRIVER ADJUSTMENTS ARE MADE BY POTENTIOMETERS LOCATED ON THE REGULATOR CARD (193X378AA G01).

1. CLIM - THE CURRENT LIMIT POTENTIOMETER ADJUSTS THE CURRENT LIMIT BETWEEN 50% AND 150% OF RATED DRIVE OUTPUT CURRENT. IT OPERATES BY REDUCING INVERTER FREQUENCY AND VOLTAGE IF THE SET CURRENT LIMIT IS EXCEEDED.

2. ATIM, DTIM - THE ACCELERATION TIME AND DECELERATION TIME POTENTIOMETERS INDEPENDENTLY ADJUST THE DRIVE ACCELERATION AND DECELERATION RATES RESPECTIVELY. FOR A 15 VOLT STEP CHANGE IN DRIVER REFERENCE, THE DRIVE WILL ACCELERATE OR DECELERATE BETWEEN ZERO AND RATED OUTPUT ON LINEAR RAMPS WHICH ARE ADJUSTABLE FROM 2.5 TO 25 SECONDS EACH.

3. VB - THE VOLTAGE BOOST POTENTIOMETER ADJUSTS THE FIXED AMOUNT OF VOLTAGE WHICH IS ADDED TO THE INVERTER IRRESPECTIVE OF FREQUENCY. THE AMOUNT OF VOLTAGE BOOST CAN BE ADJUSTED FROM ZERO TO APPROXIMATELY 7% OF RATED VOLTAGE. BELOW THE MINIMUM REFERENCE LEVEL, THE VOLTAGE BOOST IS SWITCHED OFF.

4. CLST - THE CURRENT LIMIT STABILITY POTENTIOMETER IS ADJUSTED, DEPENDING ON THE AMOUNT OF DRIVE INERTIA, TO STABILIZE THE CURRENT LIMIT FUNCTION. IT CONTROLS THE RATE AT WHICH THE CURRENT LIMIT CAN REDUCE THE MOTOR SPEED ON OVERLOAD.

5. V/HZ - THE VOLTS/HERTZ POTENTIOMETER PROVIDES A VERNIER ADJUSTMENT OF THE VOLTS PER HZ OF THE INVERTER WITHIN +15%, -5% OF NOMINAL.

6. MINF - THE MINIMUM FREQUENCY POTENTIOMETER ADJUSTS THE INVERTER MINIMUM FREQUENCY FROM 3% TO 12% OF SET BASE FREQUENCY. FOR REFERENCE LEVELS BELOW THE MINIMUM FREQUENCY LEVEL, ONLY THE INVERTER VOLTAGE IS DECREASED.

7. BF - THE BASE FREQUENCY POTENTIOMETER ADJUSTS THE INVERTER BASE FREQUENCY OVER A MINIMUM 2 TO 1 RANGE WITHIN EITHER OF THE TWO BASE FREQUENCY RANGES, 37.5 TO 75 HZ OR 75 TO 150 HZ, SELECTED BY THE BFR JUMPER (SEE NOTE C1).

AN EXTERNAL BASE FREQUENCY ADJUSTMENT MAY BE ADDED TO MODIFY THE INTERNAL SETTING BY AS MUCH AS PLUS OR MINUS 50%, AS EXPLAINED IN NOTE A7.

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REVISIONS

3	JULY 13, 1977 M.Cotty	SV-8722 #11 CHANGED NOTE 8 CER

8. VLIM (INITIAL SET-UP ADJUST) - THE VOLTAGE LIMIT POTENTIOMETER IS USED TO PREVENT SATURATION OF THE CONVERTER BY LIMITING THE REFERENCE TO THE VOLTAGE REGULATOR. IT IS ADJUSTED TO LIMIT THE MAXIMUM INVERTER A.C. OUTPUT VOLTAGE TO BE SLIGHTLY LESS THAN THE A.C. SUPPLY VOLTAGE.

CAUTION: MISADJUSTMENT OF THIS POTENTIOMETER CAN CAUSE EXCESSIVE CURRENTS DURING MOTOR TRANSFER BETWEEN LINE AND INVERTER.

F. INDICATING LIGHTS

1. ITOC (ON SYSTEM CARD) - THE INVERSE TIME OVERCURRENT TRIP LIGHT INDICATES AN INVERTER OVERCURRENT SHUTDOWN. THE DURATION OF THE OVERCURRENT CONDITION BEFORE TRIPPING IS FROM 1 MILLISECOND AT OVER 200% OF RATED CURRENT TO 1 MINUTE AT THE SET CURRENT LIMIT LEVEL.

2. IF (ON SYSTEM CARD) - THE INVERTER FREQUENCY LIGHT INDICATES THE OPERATING FREQUENCY OF THE INVERTER PHASE A CONTROL WHETHER THE INVERTER IS ACTUALLY RUNNING OR NOT. IT IS USEFUL IN DETERMINING THE MINIMUM FREQUENCY AT STANDBY.

3. SYNC (ON SYSTEM CARD) - THE INVERTER SYNCHRONIZED LIGHT INDICATES WHEN THE INVERTER IS FREQUENCY AND PHASE SYCHRONIZED TO THE A.C. LINE.

4. LOV (ON CONVERTER CARD) - THE D.C. LINK OVERVOLTAGE TRIP LIGHT INDICATES AN OVERVOLTAGE SHUTDOWN DUE TO REGENERATIVE "PUMP UP" OF THE D.C. LINK VOLTAGE.

5. PS/LOP (ON COVERTER CARD) - THE WRONG PHASE SEQUENCY/LOSS OF PHASE LIGHT WILL INDICATE AN INCORRECT PHASE SEQUENCE OF THE A.C. SUPPLY LINE CONNECTION, OR A LOSS OF ONE OR MORE OF THE A.C. SUPPLY LINE VOLTAGES.

6. 1FT (ON INVERTER CARD) - THE INVERTER FAULT TRIP LIGHT INDICATES AN INVERTER "SHOOT-THROUGH" FAULT SHUTDOWN DUE TO INVERTER MISOPERATION IF IT IS THE ONLY LIGHT INDICATION. HOWEVER, OPERATION OF THE LOV, COC AND CUV TRIP CIRCUITS WILL PRODUCE A "SHOOT-THROUGH" SHUTDOWN, SO THE 1FT LIGHT WILL ALSO INDICATE WHENEVER EITHER THE LOV, COC OR CUV LIGHTS INDICATE.

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7. IOF (ON INVERTER CARD) - THE INVERTER OVERFREQUENCY TRIP LIGHT INDICATES AN OVERFREQUENCY SHUTDOWN AS DETERMINED BY THE SELECTION OF THE FREQUENCY TRIP LEVEL JUMPERS, AS EXPLAINED IN NOTE C2.

8. COC (ON INVERTER CARD) - THE COMMUTATION OVERCURRENT TRIP LIGHT INDICATES AN OVERCURRENT SHUTDOWN DUE TO AN OVERVOLTAGE IN ANY ONE OF THE THREE INVERTER COMMUTATION CIRCUITS. THIS SHUTDOWN NORMALLY OCCURS FROM EXCESSIVE MOTOR CURRENT PEAKS AT OR NEAR BASE FREQUENCY, BUT CAN BE DUE TO COMMUTATION CIRCUIT MISOPERATION AT LOWER FREQUENCIES AND VOLTAGES.

9. CUV (ON INVERTER CARD) - THE CONTROL UNDERVOLTAGE TRIP LIGHT INDICATES A SHUTDOWN DUE TO LOW CONTROL VOLTAGE OR LOW FIRING SIGNAL VOLTAGE.

G. DRIVER READOUTS

THESE SIGNAL READOUTS, EXCEPT FOR FVRO, ARE DESIGNED TO DRIVE CARD MOUNTED RELAYS CONNECTED FROM +20V TO THE READOUT TERMINAL. THEY ARE EACH CAPABLE OF "SINKING" 15 MA IN THEIR LOW STATES.

1. OSRO - IF TB18 GOES LOW, IT SIGNALS THAT THE INVERTER IS FREQUENCY AND PHASE SYNCHRONIZED TO THE A.C. LINE.

2. 1FTRO - IF TB19 GOES HIGH, IT SIGNALS A FAULT TRIP SHUTDOWN OF THE DRIVE. THE CAUSE OF THE SHUTDOWN SHOULD BE SHOWN BY THE CARD INDICATING LIGHTS ASEXPLAINED IN NOTES F1 THROUGH F9.

3. ORRO - IF TB20 IS LOW, IT SIGNALS THAT THE INVERTER IS RUNNING. IF TB20 IS HIGH, IT SIGNALS THAT THE INVERTER IS STOPPED.

4. OMVFRO - IF TB21 IS LOW, IT SIGNALS THAT THE INVERTER IS AT ITS MINIMUM VOLTAGE AND FREQUENCY. IF TB21 IS HIGH IT SIGNALS THAT THE INVERTER IS ABOVE ITS MINIMUM VOLTAGE AND FREQUENCY.

5. FVRO - TB22 PROVIDES AN ANALOG VOLTAGE SIGNAL EQUIVALENT TO THE INVERTER FREQUENCY WITHIN ±1% OF MAXIMUM. THE INVERTER FREQUENCY WILL BE 5 TIMES THE TB22 VOLTAGE WHEN TB35 IS JUMPERED TO TB34 (SEE NOTE C1), AND WILL BE 10 TIMES THE TB22 VOLTAGE WHEN TB35 IS LEFT OPEN.

H. DIAGNOSTICS

TB36 THROUGH TB40 ARE PROVIDED FOR DIAGNOSTIC CHECKOUT OF THE DRIVE USING A GOOD QUALITY OSCILLOSCOPE HAVING APPROXIMATELY A 1 MEGOHM INPUT RESISTANCE. THESE FIVE DIAGNOSTIC TERMINALS ARE CONNECTED TO THEIR CARD SIGNALS THROUGH 15K RESISTORS TO PREVENT CONTROL MISOPERATION FROM NOISE OR OTHER SIGNALS INTRODUCED AT THESE TERMINALS. HOWEVER, EXTREME CARE SHOULD STILL BE EXERCISED WHEN USING SEL1 AND SEL2 DIAGNOSTIC TERMINALS.

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Ø1 .12" NOTE WAS X1,3 X3-4

H. DIAGNOSTICS (CONT'D)

CAUTION: CONSULT INSTRUCTION BOOK FOR PROPER PROCEDURE IN USING SEL1 AND SEL2 FOR DIAGNOSTIC CHECKOUT SINCE DRIVE MISOPERATION AND POSSIBLE DAMAGE MAY RESULT FROM INCORRECT PROCEDURE.

1. LPAD - TB36 PROVIDES A SQUARE WAVE FREQUENCY SIGNAL WHICH IS IN PHASE WITH PHASE A (OR PHASE 1) OF THE A.C. SUPPLY LINE.

2. IPAD - TB37 PROVIDES A SQUARE WAVE FREQUENCY SIGNAL WHICH IS IN PHASE WITH PHASE A OF THE INVERTER.

3. OCPD - TB38 PROVIDES A LOW GOING COMMUTATION PULSE FREQUENCY WHICH IS 6 TIMES THE INVERTER FUNDAMENTAL FREQUENCY, AND IS USEFUL AS AN OSCILLOSCOPE TRIGGER SIGNAL.

4. SEL1, SEL2 - TB39 AND TB40 ARE CONNECTED TO TWO SELECTOR PROBES WHICH CAN BE USED TO READ ANY CARD OUTPUT TERMINAL ON THE DRIVER BACK PLANE, BY MEANS OF AN OSCILLOSCOPE.

5. METER CARD - AN OPTIONAL METER CARD (193X381AA G01) CAN BE USED IN PLACE OF OR IN CONJUNCTION WITH AN OSCILLOSCOPE FOR DIAGNOSTIC CHECKOUT. CONSULT THE INSTRUCTION BOOK FOR DIAGNOSTIC PROCEDURES.

J. POWER SUPPLY INPUTS/OUTPUTS

1. H1, H2 - TB1 AND TB2 ARE FOR THE 115 VOLT A.C. SUPPLY TO THE DRIVER CONTROL POWER TRANSFORMER.

2. X1-3, X2-4 - TB3 AND TB4 ARE FOR THE 28 VOLT A.C. OUTPUT OF THE TRANSFORMER INTO THE POWER SUPPLY CARD.

3. RT30 - TB5 MAY BE CONNECTED TO A D. C. BATTERY SOURCE OF 27 TO 31 VOLTS FOR CONTROL RIDE THROUGH OF AN A.C. POWER OUTAGE.

4. PSR - TB6 MAY BE USED AS THE POWER SUPPLY REGULATING SIGNAL TO A SLAVED POWER SUPPLY.

5. +20V, COM - TB7 AND TB8 ARE THE +20 VOLT AND COMMON OUTPUTS OF THE 2.5 AMP MAXIMUM RATED POWER SUPPLY CARD.

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